Analysis

Assessment of computational methods for diabetic retinopathy lesion segmentation

Luming Yang[1,*], Siqing Li[1], Yuanchen Liu[1], Chang Yu[1], Haoxian Liu[2], Zhenyu Bu[1] and Alper Yilmaz[1,*]

[1]Photogrammetric Computer Vision Lab, The Ohio State University, Columbus, OH, USA

[2]Department of Computer Science and Engineering, The Hong Kong University of Science and Technology, Hong Kong SAR, China

*Correspondence: Luming Yang (yang.7670@osu.edu) and Alper Yilmaz (yilmaz.15@osu.edu)

Abstract

Methods for segmenting diabetic retinopathy lesions in fundus photographs are usually compared through published scores on a few benchmark datasets. We asked whether such rankings can guide the choice of a method. To answer this, we separated three properties that a published score combines: numerical reproducibility, performance under controlled conditions, and generalization across data sources. We audited the reported numbers of 29 method papers, rescored and reran released models, retrained 19 models under one protocol, and tested them on eight test sets from seven datasets. Reported numbers depended on the evaluation implementation. A widely reused fixed-threshold AUPR inflated scores by model-dependent amounts and changed the order of models, and fewer than a third of baseline values had been rerun by the papers that report them. Under controlled conditions, published rankings did not predict performance, and the leading models could not be separated from one another. Across sources, scores and rankings held only where annotation conventions and imaging conditions matched those of the training data. Comparisons of lesion segmentation methods should therefore report these three properties separately. We will release the code, the splits, the probability maps and a reference model as a benchmark, RetiSeg[†].

Diabetic retinopathy (DR) is a leading cause of vision loss in working-age adults; about 103 million adults had DR in 2020, and 160.5 million are projected to have it by 2045 [1,2]. Screening programmes therefore photograph the fundus of people with diabetes at regular intervals and grade the images from visible lesions, chiefly microaneurysms (MA), hemorrhages (HE), hard exudates (EX) and soft exudates or cotton wool spots (SE)[‡]. The number, type and location of these lesions define the severity scales and the decision to refer [3,4]. Deep networks now detect referable DR at the image level with high sensitivity and specificity [5,6], and an autonomous system has been authorized for use in primary care after a prospective pivotal trial [7]. Such classifiers do not show why an image was referred, however. Their sensitivity also varies widely in real-world screening [8], and the reference standard against which they are judged depends on the graders who produce it [9].

Lesion segmentation addresses these limits directly. A model that outlines lesions shows a grader why an image received its grade, supports screening decisions that must be auditable, and provides quantitative lesion burden for follow-up [10–12]. Public datasets with pixel-level labels, first IDRiD [13,14] and DDR [15] and later FGADR [16], Retinal-Lesions [17], TJDR [18] and MAPLES-DR [19], have made the task a standard benchmark. New methods, from task-specific networks to adapted foundation models, are routinely ranked by their scores on IDRiD and DDR. Whether these rankings can guide the choice of a method depends on three properties that published comparisons rarely separate: whether a reported number can be reproduced, whether a difference persists when training and evaluation are fixed, and whether it carries over to data from other sources.

[†]The RetiSeg code repository is at https://github.com/skylynf/retiseg and is still being refined; the probability maps and the weights of the reference model will be released there.

[‡]The lesion terminology of this version follows the public datasets. We are reviewing these terms with clinicians and plan to revise them in a later version.

Each of these properties depends on factors outside the model. (1) The evaluation implementation. Papers differ in the split, the resolution at which predictions are scored, whether pixels outside the field of view count, and how the area under the precision–recall curve (AUPR) is computed. Such choices are known to reorder methods[20,21]. Baseline numbers are also carried between papers without being rerun, and overlap between training, pretraining and test images, a common source of optimistic results[22], has to be checked for each source. (2) The annotation conventions. Sources were labelled with polygons, ellipses or regions, by many graders or by a few experts[23], so "ground truth" differs from one source to the next. A test split drawn from the training source therefore says little about performance elsewhere[24]. (3) The imaging conditions. The lesions are small: shrinking an image to the 224-pixel input of a common backbone removes 34.3% of the microaneurysms from the DDR training labels. Fields of view also range from standard to ultra-widefield.

Earlier syntheses do not resolve these issues. Surveys collect the numbers that papers report[10,11]. The closest study[25] showed with U-Net-type baselines that annotation style governs transfer between datasets, but it did not rerun fundus-specific methods, audit published numbers or include foundation models. Outside ophthalmology, standardized reassessments have found that part of the reported progress disappears once baselines are configured with care[26,27]. A recent assessment of T cell receptor–epitope predictors[28] supplies the design we follow: evaluate the original models, retrain everything that can be retrained under one protocol, and judge both on independent data.

We assessed DR lesion segmentation methods at each of the three levels. For numerical reproducibility, we audited the published numbers and their provenance, scored the released checkpoints and reran the authors' pipelines. For performance under controlled conditions, we retrained 19 models under one protocol on two sources. For generalization, we tested every model on six test sets from five sources never used for training, and varied the training source, annotation granularity, input resolution and training amount one at a time. The findings are distilled into a reference model, RetiSeg-Ref. Because pixel-level labels are themselves graders' judgements, we are also reviewing model errors with ophthalmologists; this clinical review will be reported in a later version.

Results

Study design

We separated three properties that a published score combines: whether the number can be reproduced, how the model performs when training and evaluation are fixed, and whether its performance carries over to data from other sources (Fig. 1a). Eight test sets were prepared from seven datasets, 5,121 images in total (Fig. 1b and Table 1). IDRiD and DDR are the training sources, because both have official splits and almost every published number refers to one of them. No model in the main tables was trained on the other six test sets, which we call independent. The sources also differ in how lesions were drawn. Following Playout and Cheriet[25], we assigned each source to one of four annotation granularity tiers: fine, mixed, coarse and region consensus.

From a ledger of 86 records (Fig. 1d; Methods) we registered 39 model variants in five families. Each has a reproducibility level from R0 (no code) to R4 (training code, weights, splits and environment) (Fig. 1c and Table 2). Tier 1 holds the 22 variants of the main comparisons: general segmentation networks (U-Net[29], DeepLabv3[30,31], HRNet[32], Swin-Unet[33], FCT[34] and H2Former[35]), the self-configuring nnU-Net[26], seven fundus-specific networks (M2MRF[36], HACDR-Net[37], HRDecoder[38], PMCNet[39], WSRFNet[40], CLC-Net[41] and TC-Net[42]) and eight foundation models (SAM[43], SAM 2[44], SAM 3[45], MedSAM[46], SAMed[47], SAM-Adapter[48], SAM2-UNet[49] and RETFound[50]). H2Former exceeds the memory of one card at the unified patch

Table 1 Datasets, frozen splits and roles. All eight sources carry pixel- or region-level labels for microaneurysms (MA), hemorrhages (HE), hard exudates (EX) and soft exudates (SE). Granularity decides which test sets may be averaged: averages are taken only within one tier. Split is train / validation / test in images; official test sets are never changed. "Maps" marks sources whose probability maps we can release.

Source	Ref.	Images	Resolution	FOV	Annotation	Granularity	Split	Maps
Training sources (internal test = official test set)								
IDRiD	[13]	81	4288×2848	50°	pixel contours, expert review	Fine	44 / 10 / 27	✓
DDR	[15]	757	variable	45°	pixel masks	Fine	383 / 149 / 225	✓
Independent test sources (never used to train a main-table model)								
TJDR, standard field	[18]	257	2048×2048	35–50°	polygons, 2 graders + adjudicator	Fine	172 / 30 / 55	✓
TJDR, ultra-widefield	[18]	304	3912×3912	133°	polygons, 2 graders + adjudicator	Fine	209 / 37 / 58	✓
MAPLES-DR	[19]	198	1440×960–2304×1536	45°	pixel masks, 7 retinologists; MA/HE/EX AI-assisted	Fine	– / 0 / 60	✓
FGADR	[16]	1,842	1280×1280	n.s.	pixel masks, 2 residents + senior	Mixed	1,096 / 193 / 553	✗
Retinal-Lesions	[17]	1,593	896×896	n.s.	ellipses and polygons, ≥3 of 45 graders	Coarse	951 / 164 / 478	✗
DiaRetDB1	[23]	89	1500×1152	50°	regions with confidence, 4 experts	Region	– / 28 / 61	✓

FOV, field of view; n.s., not stated in the source paper. IDRiD: 10 of the 54 official training images are held out for validation in unified retraining; author-recipe retraining uses all 54. DiaRetDB1: ground truth is the mean confidence of four experts thresholded at 0.75; the 28 official training images are used only to choose thresholds. Retinal-Lesions: split by patient; grey-level 127 regions are ignored. MAPLES-DR: labels drawn at 1500 × 1500 on the field-of-view crop of MESSIDOR images and placed back at the original resolution; the 138 training images are not used, the official 60 test images are; uncertain red (MA or HE) and bright (EX, SE or drusen) lesions are ignored for the two classes they could belong to. The FGADR licence forbids redistribution of derived data; maps for Retinal-Lesions are not released either.

size and is deferred. Tier 2 (6 variants, such as L-Seg[51] and RTNet[52]) is registered for later rounds, and tier 3 (11 variants, including semi-supervised and text-prompted methods[53–55]) is audited from the papers only.

Each property is examined with its own evaluations, and their results are never mixed in one table column. Numerical reproducibility is examined with the paper audit, the released checkpoints (author weights) and reruns of the authors' pipelines (author recipes). Performance under controlled conditions is examined by retraining the tier-1 models under one protocol, the unified protocol, and testing them on the official test set of their training source (internal test). Generalization is examined by applying every checkpoint unchanged to the independent test sets, and by varying the training source, annotation granularity, input resolution and training amount one at a time (Fig. 1e). The primary metric is the exact AUPR of each lesion class over all field-of-view pixels of a test set, averaged over the four classes (mAUPR).

Published numbers are not comparable across papers

Few methods can be rerun at all. Of the 21 fundus-specific methods in the registry, 6 have author code, 4 have training code, and 2 come with weights, splits and an environment description (Fig. 2a).

Papers state their split but not their metric. We extracted the setup and comparison tables of 29 method papers, 1,538 baseline AUPR values on IDRiD and DDR in total (Supplementary Tables 1 and 2). Most papers use the official split: 27 of the 30 papers on IDRiD and 20 of the 24 on DDR. Of the 27 papers whose main metric is AUPR, 18 do not say how it was computed, and 6 integrate over a fixed grid of thresholds, the IDRiD challenge protocol carried on by the evaluation code released with L-Seg and M2MRF. No paper states whether pixels outside the field of view were excluded (Fig. 2b).

Baseline numbers are mostly carried from earlier papers. Only 28% of the baseline values were produced by the paper that reports them. Another 22% repeat the value in the method's own paper and 26% repeat a value that an earlier paper had already carried; 23% cite a source outside the ledger or none. When a method is rerun, the value changes: L-Seg is reported on IDRiD with mAUPR from 0.40 to 0.66 across 14 papers, and U-Net from 0.35 to 0.65 across 6 (Fig. 2c).

The evaluation implementation changes scores and ranks

The widely reused 11-threshold implementation inflates AUPR. Its lowest threshold is 0, at which every pixel with a non-zero probability counts as positive. This adds a point with a recall of 1 and a precision equal to the prevalence, which enlarges the area under the curve. For the released M2MRF checkpoint, the authors'

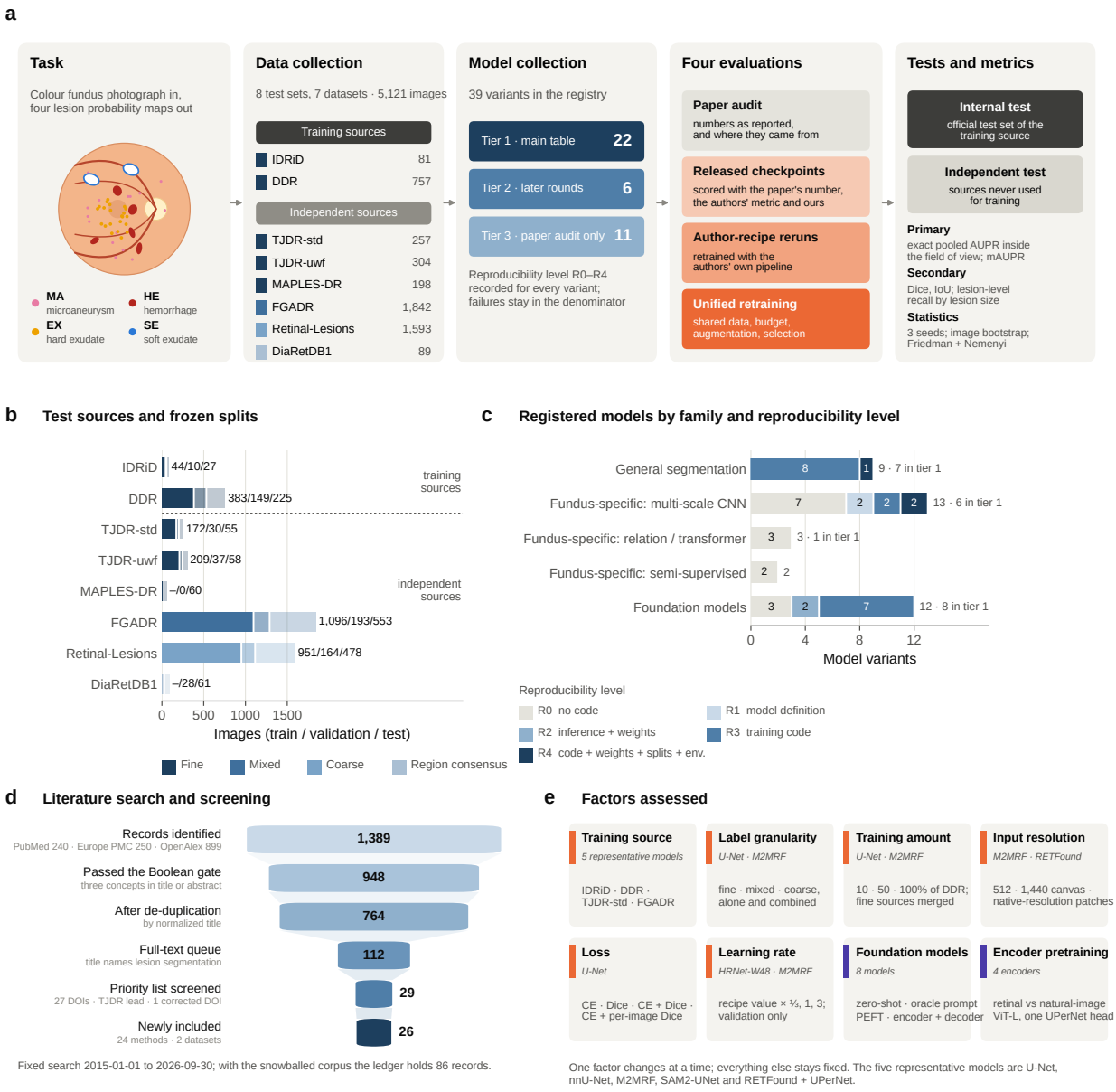


Fig. 1 Data collection and study design. a, Workflow: the task, the eight test sets, the registered model variants, the four kinds of evaluation, and the two test scenarios with their metrics. **b**, Images per source and frozen split (train / validation / test). Colour gives the annotation granularity tier; shade separates the three parts of a split. **c**, Registered model variants by family and reproducibility level; the number of tier-1 variants is written after each bar. **d**, Literature search and screening. **e**, Factors varied one at a time, with the models used for each.

Table 2 Models entering the retraining tracks. 39 variants are registered: 22 in tier 1 (main tables), 6 in tier 2 (later rounds) and 11 in tier 3 (paper audit only; Supplementary Note 1). Reproducibility level: R0, no code; R1, model definition only; R2, inference code and weights; R3, training code; R4, training code, weights, splits and environment. Tracks: A, author weights; B, unified retraining of the official code; C, re-implementation from the paper, marked [C] in the tables.

			Track				
Model	**Tier**	**Reproducibility**	**A**	**B**	**C**	**Code status**	**Use in this study**
General segmentation and self-configuring baselines							
U-Net	1	●●●○ R3	✗	✓	✗	reference implementation	unified retraining
DeepLabv3	1	●●●○ R3	✗	✓	✗	torchvision, ResNet-101	unified retraining
HRNet-W48	1	●●●○ R3	✗	✓	✗	official, W48	unified retraining
nnU-Net	1	●●●● R4	✗	✓	✗	official, 2D; default and ResEnc-M	own entry point
Swin-Unet	1	●●●○ R3	✗	✓	✗	official	unified retraining, 224 patches
FCT	1	●●●○ R3	✗	✓	✗	official PyTorch port	unified retraining, 384 patches
H2Former	1	●●●○ R3	✗	✓	✗	official	deferred (memory at the unified patch size)
UNet++	2	●●●○ R3	✗	✓	✗	official	tier 2
UPerNet	2	●●●○ R3	✗	✓	✗	official	tier 2
Fundus-specific, multi-scale CNN							
M2MRF	1	●●●● R4	✓	✓	✗	official; weights released	released weights; author-recipe rerun; unified retraining
HACDR-Net	1	●●●○ R3	✗	✓	✗	official; no weights	unified retraining
HRDecoder	1	●●●● R4	✓	✓	✗	official; weights released	released weights; author-recipe rerun; unified retraining
WSRFNet	1	●○○○ R1	✓	✓	✗	model and test code only; training loop added	unified retraining
PMCNet	1	●●●○ R3	✗	✓	✗	official Keras code rewritten in PyTorch	unified retraining
CLC-Net	1	○○○○ R0	✗	✗	✓	no code; re-implemented from the paper	unified retraining (re-implementation)
L-Seg	2	●○○○ R1	✗	✗	✓	Caffe prototxt and loss layer	C
SAA	2	○○○○ R0	✗	✗	✓	no usable code	C
Fundus-specific, relation or transformer							
TC-Net	1	○○○○ R0	✗	✗	✓	no code; re-implemented from the paper	unified retraining (re-implementation)
RTNet	2	○○○○ R0	✗	✗	✓	no code; image + vessel input	C
Foundation models							
SAM	1	●●○○ R2	✗	✓	✗	official inference	zero-shot, oracle prompt
SAM 2	1	●●●○ R3	✗	✓	✗	official	zero-shot, oracle prompt
SAM 3	1	●●●○ R3	✗	✓	✗	official; gated weights	text prompt
MedSAM	1	●●●○ R3	✗	✓	✗	official	oracle box, full fine-tuning
SAMed	1	●●●○ R3	✗	✓	✗	official	PEFT (LoRA)
SAM-Adapter	1	●●●○ R3	✗	✓	✗	official	PEFT
SAM2-UNet	1	●●●○ R3	✗	✓	✗	official	PEFT
RETFound + UPerNet	1	●●●○ R3	✗	✓	✗	official encoder; UPerNet head added	encoder + decoder
PerSAM	2	●●○○ R2	✗	✓	✗	official inference	one-shot, tier 2

general segmentation self-configuring fundus-specific foundation model. Unified retraining, the shared protocol (Methods). PEFT, parameter-efficient fine-tuning. Code status as of the audit of 2–4 October 2026.

function returns 0.676 on IDRiD and 0.492 on DDR. The exact AUPR inside the field of view is 0.639 and 0.401 (Fig. 2d and Table 3). The authors' function reproduced their repository values to the last printed digit, and the field of view and overlapping labels changed only the third decimal (Fig. 3a,b). The gap between paper and protocol is therefore a property of the metric.

The inflation differs between models, so it changes their order. Applied to the maps of the converged retrained models, the 11-threshold implementation changed mAUPR on IDRiD by −0.010 to +0.023. On DDR, where the low-probability tail of the maps carries more weight, it added 0.004 to 0.084 (Supplementary Table 5). The largest additions went to the least calibrated maps, and 10 of the 19 models changed rank (Kendall $\tau = 0.88$; Fig. 2e).

Released training code did not reproduce the released weights. We reran the M2MRF pipeline on IDRiD with the authors' batch size and schedule, three times. With the authors' metric, the three runs reached 0.605–0.627, against 0.676 for the released checkpoint (Fig. 3c). Every run was 5–7 points lower, about five times the spread between runs. HRDecoder, rerun once with its authors' recipe, reached 0.674 under our protocol, 0.03 below its released checkpoint (0.707).

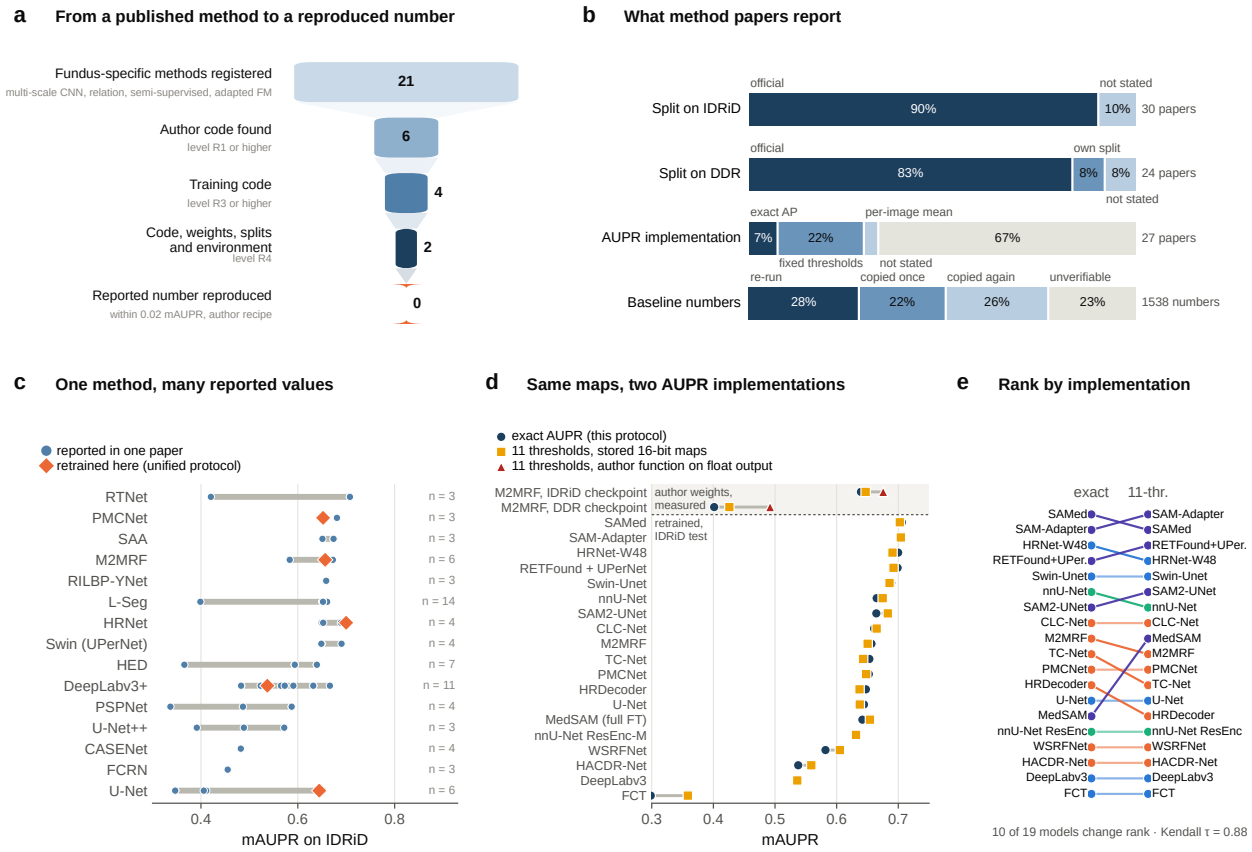


Fig. 2 The numbers in the literature. **a**, From registered fundus-specific methods to a reproduced number; the first four stages follow the reproducibility levels of the registry. **b**, Share of method papers by split and by AUPR implementation (fixed thresholds, a grid of 11 or 33 operating points), and provenance of the baseline AUPR values on IDRiD and DDR in their comparison tables: produced by the reporting paper, identical to the method's own paper, identical to a value already carried by an earlier paper, or citing a source outside the ledger. **c**, mAUPR on IDRiD reported for one method in different papers (dots; bar, range; n, number of papers; methods reported by at least three papers) and the value after unified retraining (diamond). **d**, The same probability maps scored with the exact AUPR and with the 11-threshold implementation; the two shaded rows are the released M2MRF checkpoints, for which the authors' function on the float output is also shown. **e**, Rank of the retrained models under the two implementations.

Under controlled conditions the leading models cannot be separated

For a controlled comparison we fixed everything that is not part of a method: data and splits, a canvas of 1,440 pixels across the field of view, the training amount counted in pixels, augmentation, checkpoint selection by validation mAUPR, inference geometry and evaluation (Methods). Each model keeps its own loss, optimizer, learning rate, weight decay, schedule shape and effective batch (Table 8). Source recipes could not serve as common ground, because moved to IDRiD unchanged they differ about 67-fold in the number of training images seen. We retrained 19 models on IDRiD and on DDR, nine of them with three seeds (Supplementary Table 3). Tables and tests use the mean over seeds.

Two adapted foundation models and one general network led both internal tests: SAMed, RETFound with a UPerNet head[56] and HRNet-W48 (Fig. 4a and Table 4). SAMed is a low-rank adaptation[57] of the SAM ViT-B[58] encoder. On IDRiD, SAM-Adapter scored level with SAMed. The differences among the leading models were within their variation between seeds. On DDR, SAMed and RETFound were 0.013 and

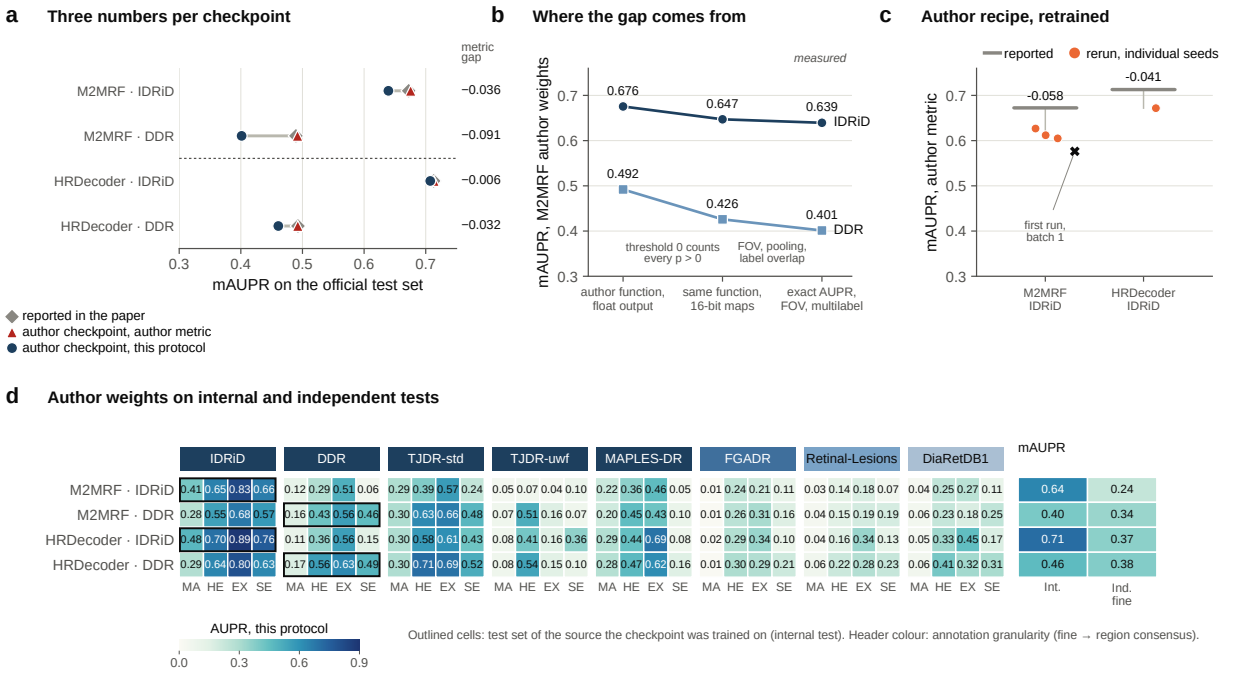


Fig. 3 Performance of the original models. a, mAUPR of each released checkpoint on the official test set of its training source: reported in the paper, authors' function on the checkpoint, and this study's protocol. The number on the right is the difference between the last two. **b**, Decomposition of that difference for the two M2MRF checkpoints. **c**, Retraining with the authors' recipe: reported value (line) and individual seeds (dots), with the mean difference above; the cross is the first rerun at an effective batch of 1. **d**, Per-class AUPR of the author weights on all eight test sets under this study's protocol. Outlined cells are internal tests.

0.004 above HRNet-W48, and their own seed standard deviations were 0.027 and 0.018. Their 95% bootstrap intervals over test images overlap (Supplementary Table 4). The table as a whole spans 0.41 mAUPR on IDRiD and 0.28 on DDR. IDRiD has only 27 test images, so we base comparisons between architectures on DDR and on the independent tests.

Lesion-specific designs did not lead. The best fundus-specific network on DDR, CLC-Net, was 0.05 below SAMed. M2MRF and HRDecoder add lesion-specific decoders to the HRNet-W48 backbone, and both scored below the general HRNet-W48 head on both sources. The released HRDecoder checkpoints score higher than its unified run; a case study traces this difference to parts of the authors' recipe that the unified protocol replaces (Supplementary Note 2).

Published rankings did not predict the controlled ranking. For the 10 models that report an IDRiD mAUPR, the published order and the order after retraining were unrelated (Kendall $\tau = -0.11$, $P = 0.73$; Fig. 4f).

The largest departures came from recipe components. DeepLabv3 decodes at an output stride of 8 and lost the smallest class (microaneurysm AUPR 0.043 on DDR). FCT, trained from scratch at a constant learning rate, did not converge within the budget. A Dice term computed per image drove the rare-class channels to zero on DDR, where 83% of the training patches contain no soft exudate. This affected the registered recipes of U-Net, SAM2-UNet and MedSAM. Pooling the Dice term over the batch raised their DDR mAUPR from 0.119, 0.210 and 0.265 to 0.378, 0.360 and 0.413 (seed 0). For SAM2-UNet and MedSAM this revision followed a rule registered before training; for U-Net it was made after the registered run had been scored (Methods). Table 4 reports the revised runs, and Supplementary Table 6 the registered ones. Other choices mattered less. With cross-entropy alone instead of cross-entropy and a pooled Dice term, U-Net lost 0.02 on

Table 3 Original models: three numbers per released checkpoint, and retraining with the authors' recipe. a, Each checkpoint is scored on the official test set of its training source with the number printed in the paper, the authors' evaluation function (11 thresholds, whole image, overwritten labels) on the float output, the same function on the stored 16-bit maps, and this study's protocol (exact pooled AUPR inside the field of view, overlapping labels); under each per-class value is its 95% bootstrap interval over test images. **b**, The authors' pipeline rerun from scratch; last iteration, no checkpoint selection.

a Released checkpoints (author weights)

		mAUPR, author metric			AUPR, this protocol						
Model	**Source**	Paper	Function	Stored	**mAUPR**	MA	HE	EX	SE	**Δ metric**	**Indep. fine**
M2MRF	IDRiD	.672	.676	.647	.639	.408 .35–.42	.654 .57–.66	.832 .72–.85	.664 .44–.76	▼ .036	.236
M2MRF	DDR	.489	.492	.426	.401	.165 .12–.20	.426 .28–.48	.558 .47–.61	.456 .29–.55	▼ .091	.341
HRDecoder	IDRiD	.713	.713	.715	.707	.475 .44–.51	.699 .65–.74	.892 .81–.93	.763 .58–.87	▼ .006	.369
HRDecoder	DDR	.493	.492	.492	.461	.172 .13–.22	.557 .51–.60	.629 .56–.70	.486 .33–.60	▼ .032	.384

b Retrained with the authors' recipe (author-recipe reruns)

Model	**Source**	**Paper**	**Retrained, author metric**	**Seeds**	**Δ to paper**	**Retrained, this protocol**
M2MRF	IDRiD	.672	.615 ± .011	3	▼ .058	.552 ± .013
first run, batch 1	IDRiD	.672	.576	1	▼ .096	.493
HRDecoder	IDRiD	.713	.672	1	▼ .041	.674

"Δ metric" is this protocol minus the authors' function on the same checkpoint. "Indep. fine" is the mean mAUPR on the three fine-grained independent test sets (TJDR standard-field, TJDR ultra-widefield and MAPLES-DR). The M2MRF paper reports the mean of three runs (67.24 on IDRiD, 48.94 on DDR), the repository the best run (67.55, 49.20). HRDecoder: the released checkpoints (repository table 71.34, 49.24) through the authors' test pipeline (resize to 2880 × 1920 with sliding 1920 windows on IDRiD, 2048 × 2048 whole image on DDR). The first M2MRF rerun used an effective batch of 1 instead of the authors' 4.

DDR (Extended Data Fig. 2). A threefold change of the learning rate reordered HRNet-W48 and M2MRF on validation only at the lower rate (Extended Data Fig. 1).

Foundation models depend on how they are used

Results for segmentation foundation models are often reported without stating how the model was used. We separated four uses and do not rank them against each other (Fig. 5a and Table 5): zero-shot use, prompts derived from the ground truth, parameter-efficient fine-tuning, and an encoder with a trained decoder or full fine-tuning.

Without training, the models did not find lesions. The automatic masks of SAM and SAM 2 reached an any-lesion AUPR of at most 0.08 on IDRiD, against 0.78 for nnU-Net. Text prompts to SAM 3 found hard exudates and hemorrhages but not microaneurysms, and the wording of the prompt changed the result as much as the lesion class (Fig. 5c). Given a box around every lesion, by contrast, SAM delineated lesions as well as the supervised models within the same boxes (0.86 against 0.86 for nnU-Net on IDRiD). The deficit of the promptable models lies in finding lesions, not in drawing them (Supplementary Note 2).

Once trained on the task, the adaptations competed with the best supervised networks, as described above. Among them, the parameter-efficient SAMed was the strongest, and full fine-tuning of MedSAM did not exceed it.

Retinal pretraining did not improve the encoder. We compared two retinal encoders with their own starting

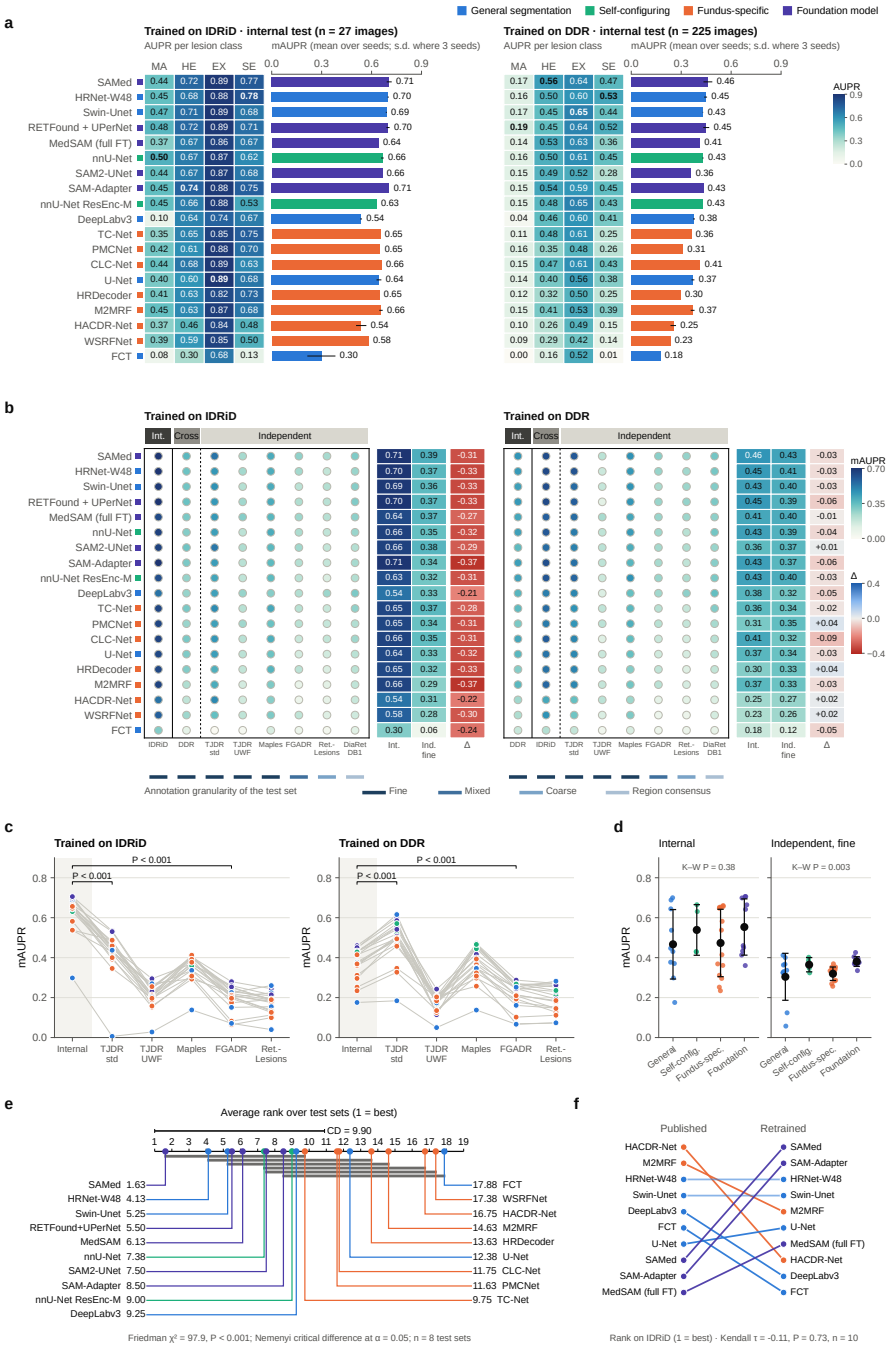


Fig. 4 Performance of retrained models under the unified protocol. a, Internal tests: per-class AUPR (mean over seeds) and mAUPR (bar; error bar, s.d. over seeds where three were run). Models are ordered by their average rank in **e**; the square gives the family. **b**, mAUPR of every model on the eight test sets, by training source. Int., internal test; Cross, test set of the other training source; the bar under each column gives the annotation granularity. Columns on the right: internal test, mean of the three fine-grained independent test sets, and their difference. **c**, The same models on the internal test and on four independent test sets; each line is one model. P values, two-sided Wilcoxon signed-rank tests across models with Holm correction (n = 19). **d**, mAUPR by family on internal and on fine-grained independent tests; coloured dots, one model and training source; black, mean ± s.d.; K–W, Kruskal–Wallis test. **e**, Critical-difference diagram over the eight test sets; bars join models whose average ranks differ by less than the Nemenyi critical difference. **f**, Rank on IDRiD as published and after retraining.

points under one UPerNet head, one recipe and one training amount (Fig. 5d; Supplementary Table 10). On DDR, RETFound's masked-autoencoder encoder scored 0.053 below the ImageNet masked autoencoder[59] it was initialised from. RETFound's DINOv2 encoder scored 0.020 below DINOv2[60]. Both differences had the same sign on IDRiD. The two natural-image encoders, run as single-seed controls, exceeded every model of Table 4 on both sources.

Across sources, rankings hold only where annotations are drawn alike

A new source did not by itself cost performance. The DDR-trained models scored higher on IDRiD and on the standard-field TJDR set than on their own test set, and the same on MAPLES-DR (Fig. 4b,c and Table 6). These three sets share fine-grained pixel labels and a standard field of view.

Different annotation conventions and imaging fields did cost performance. The same models lost a third to a half of their internal score on the mixed and coarse labels of FGADR and Retinal-Lesions, on the region consensus of DiaRetDB1 and on the 133° ultra-widefield TJDR images. Every model was lower on each of these four sets. The models trained on the 44 IDRiD images scored higher internally but lower on every other source, including the fine-grained ones. On the six independent sets, the DDR-trained checkpoint of a model was better than its IDRiD-trained checkpoint in 74 of 114 pairs. The released author checkpoints showed the same pattern (Fig. 3d).

Internal rank predicted independent rank to the extent that labels and imaging matched. For DDR-trained models, the Spearman correlation between the internal score and the score on the three fine-grained sets was 0.82–0.92. On the mixed, coarse and region-labelled sets it was 0.76–0.79, and DeepLabv3, tenth internally, ranked first on Retinal-Lesions and DiaRetDB1, because its output stride of 8 yields region-sized predictions. On the ultra-widefield images the correlation vanished ($\rho = 0.01$). Averaged by family, the adapted foundation models were ahead both internally and on the fine-grained independent sets, and the fundus-specific and general networks were level (Fig. 4d).

Over all eight test sets, 11 of the 19 models could not be separated from the first. The Friedman test rejected equal ranks ($\chi^2 = 97.9$, $P < 0.001$), and SAMed had the best average rank, but the Nemenyi critical difference with eight blocks is 9.90 ranks (Fig. 4e). RETFound's pretraining corpus contains the EyePACS images from which Retinal-Lesions and part of FGADR are drawn (Methods); without these two sets its rank is unchanged (fourth).

Absolute scores also depend on the disease stage of the test images. On the DDR test set, the median mAUPR of the DDR-trained models was 0.21 on images of grades 0–1 and 0.36 on images of grades 2–4 (Supplementary Note 2). Scores are therefore comparable within a test set, not across test sets.

Annotation conventions decide which training data transfer

The sources differ in how lesions were outlined (Fig. 6a; Methods). In IDRiD and DDR, lesions are drawn as many small components, with a median diameter of about 0.5% of the field-of-view width. In Retinal-Lesions and DiaRetDB1, components are few and eight to nine times larger for hard exudates, so one outline encloses several lesions and the tissue between them. FGADR lies between the two.

The labels of the test set, not the size of the training set, decided which source transferred best. We trained five representative models on two further sources, the standard-field TJDR images and FGADR, at the pixel budget of DDR (Fig. 6b–d). Every source served its own test set best, and training elsewhere cost 0.15 mAUPR on average. DDR (383 images) was the best foreign source on every standard-field test set with pixel-level labels. FGADR is three times larger but labelled as mixed regions. It was the best foreign source only on the coarse and region-labelled sets and on the ultra-widefield images.

Coarse labels added to fine ones did not help. At a fixed number of patch pixels, we added Retinal-Lesions or FGADR images to DDR training for U-Net and M2MRF (Fig. 6e and Table 7). Adding Retinal-Lesions

lowered M2MRF by 0.04–0.19 on every standard-field fine-grained test set. Adding FGADR lowered M2MRF by 0.01–0.06, and moved U-Net by −0.01 to +0.05.

Correcting individual masks within one annotation style changed little. A refined annotation of the IDRiD test images has been published[61]. Scored against it, the mAUPR of the 19 IDRiD-trained models changed by at most 0.002, and their order did not change (Supplementary Note 2 and Supplementary Table 9).

Imaging conditions limit the detection of small lesions

Lesion classes differed more than models. The median AUPR over the 19 models was 0.87 for hard exudates and 0.44 for microaneurysms on IDRiD, and 0.60 and 0.15 on DDR (Supplementary Table 4). Lesion size is the main reason (Fig. 7a). Resizing a whole image to a small input removes lesions from the training target itself. At 224 pixels, 34.3% of the microaneurysms in the DDR training labels no longer cover a single pixel; on the 1,440-pixel canvas essentially none are lost (Fig. 7b).

Lesion-level recall followed size for every model (Fig. 7c). On the DDR test set, the median recall over models was 0.26 for small, 0.49 for medium and 0.66 for large lesions. Input resolution acted through the small lesions. Shrinking the canvas from 1,440 to 512 pixels lowered DDR mAUPR by 0.03 for M2MRF and RETFound with UPerNet, and their recall of small lesions by 37–64% (Fig. 7d). Patches at native resolution raised small-lesion recall but lowered mAUPR; for RETFound, the two smaller classes gained and the two larger classes lost.

The field of view had a larger effect than the inference scale. On the ultra-widefield images, the DDR-trained models changed by −53% relative to their internal score on average. Resizing these images to the 1,440-pixel canvas, instead of predicting at native resolution, moved the median mAUPR by only +0.02 (Extended Data Fig. 3b).

More fine-grained images helped, with modest gains outside the training sources. Both U-Net and M2MRF improved from 10% to 100% of the DDR training set at an equal pixel budget (Fig. 7e). Merging the three fine-grained sources (599 images) changed the DDR score by less than 0.02 and raised the scores on sets outside the training sources by 0.00–0.04.

Ensembles, a reference model and cost

Averaging the maps of several retrained models gave the highest scores of the study. The gain over the best single model was 0.02–0.06 mAUPR on the fine-grained test sets and −0.01 to 0.02 on the others (Table 6, lower block). A reference model with two members, RetiSeg-Ref, combines RETFound with UPerNet and SAMed trained on the merged fine-grained sources. On the four standard-field independent sets it came within 0.02 of the ensemble of all 18 DDR-trained models (Supplementary Table 11).

Accuracy was not related to training cost, inference cost or parameter count (Extended Data Fig. 4). HRNet-W48 matched the two leading foundation models at a twentieth of the inference time of RETFound with UPerNet. Further results on ensembles, cost and agreement with human experts are given in Supplementary Note 2.

Table 4 Unified retraining: internal tests. Every model is trained on one source under the shared protocol and scored on that source's official test set. Per-class values are exact pooled AUPR inside the field of view, averaged over seeds; mAUPR is the mean of the four classes ± s.d. over seeds for the models trained with three seeds (no s.d. is printed for the single-seed rows; Supplementary Table 3 lists every run); Dice is pooled, at the per-class threshold chosen on the validation set. 95% bootstrap intervals over test images are given per class and model in Supplementary Table 4. Cell shading follows AUPR. **Bold**, best in column; underlined, second.

		Trained and tested on IDRiD (44 / 10 / 27)						**Trained and tested on DDR** (383 / 149 / 225)					
Model	**Params**	MA	HE	EX	SE	**mAUPR**	**Dice**	MA	HE	EX	SE	**mAUPR**	**Dice**
General segmentation													
U-Net	7.8	.403	.595	**.894**	.685	.644 ± .015	.622	.137	.405	.562	.380	.371 ± .009	.405
DeepLabv3	58.6	.101	.641	.739	.668	.537 ± .006	.515	.043	.460	.600	.405	.377 ± .005	.380
HRNet-W48	65.8	.449	.682	.884	**.784**	.700 ± .004	.627	.156	.500	.597	**.535**	.447 ± .009	.448
Swin-Unet	27.2	.470	.706	.894	.683	.688 ± .003	.640	.174	.455	**.654**	.439	.431 ± .001	.440
FCT	1.9	.081	.304	.680	.128	.298 ± .084	.330	.001	.163	.524	.014	.176	.197
Self-configuring													
nnU-Net	59.2	**.499**	.665	.872	.624	.665 ± .008	.656	.157	.503	.614	.449	.431 ± .003	.458
nnU-Net ResEnc-M	105.9	.449	.663	.878	.533	.631	.638	.147	.481	.653	.432	.428	**.463**
Fundus-specific													
M2MRF	70.3	.450	.628	.867	.681	.657 ± .010	.648	.148	.406	.534	.386	.369 ± .011	.407
HACDR-Net	24.6	.366	.463	.837	.484	.538 ± .031	.520	.105	.263	.487	.155	.252 ± .017	.289
HRDecoder	65.6	.410	.629	.823	.728	.647	.636	.124	.318	.496	.245	.296	.373
PMCNet	12.9	.418	.610	.883	.698	.652	.616	.157	.346	.482	.262	.312	.338
WSRFNet	9.7	.388	.588	.849	.502	.582	.641	.088	.292	.422	.135	.234	.341
CLC-Net^{C}	119.7	.437	.683	.889	.634	.661	.632	.152	.466	.612	.428	.414	.462
TC-Net^{C}	14.9	.355	.650	.854	.754	.653	.607	.112	.476	.607	.250	.361	.352
Foundation models, adapted													
SAMed	91.6	.439	.724	.889	.772	**.706** ± .015	.658	.172	**.559**	.638	.471	**.460** ± .027	.454
SAM-Adapter	641.3	.451	**.740**	.882	.750	.706	**.663**	.147	.538	.592	.454	.433	.457
SAM2-UNet	216.5	.439	.670	.871	.678	.664	.653	.153	.493	.518	.278	.360	.413
MedSAM (full FT)	93.7	.373	.665	.855	.674	.642	.609	.136	.529	.626	.361	.413	.432
RETFound + UPerNet	352.6	.480	.721	.889	.706	.699 ± .009	.650	**.193**	.453	.641	.516	.450 ± .018	.447

Params in millions. nnU-Net uses its own preprocessing and planning with the same splits, test images and evaluation. nnU-Net plans its network per dataset: the default configuration has 59.2 M parameters on both sources, the ResEnc-M preset 105.9 M under its IDRiD plan (printed) and 140.0 M under its DDR plan. ^{C}, no author code: re-implemented from the paper (Track C); the row shows what the published description yields under this protocol, not the authors' implementation. HRDecoder, M2MRF and HRNet-W48 share the HRNet-W48 backbone. H2Former is registered in tier 1 but deferred: at its 960-pixel input one training step exceeds the 40 GB of the cards used here.

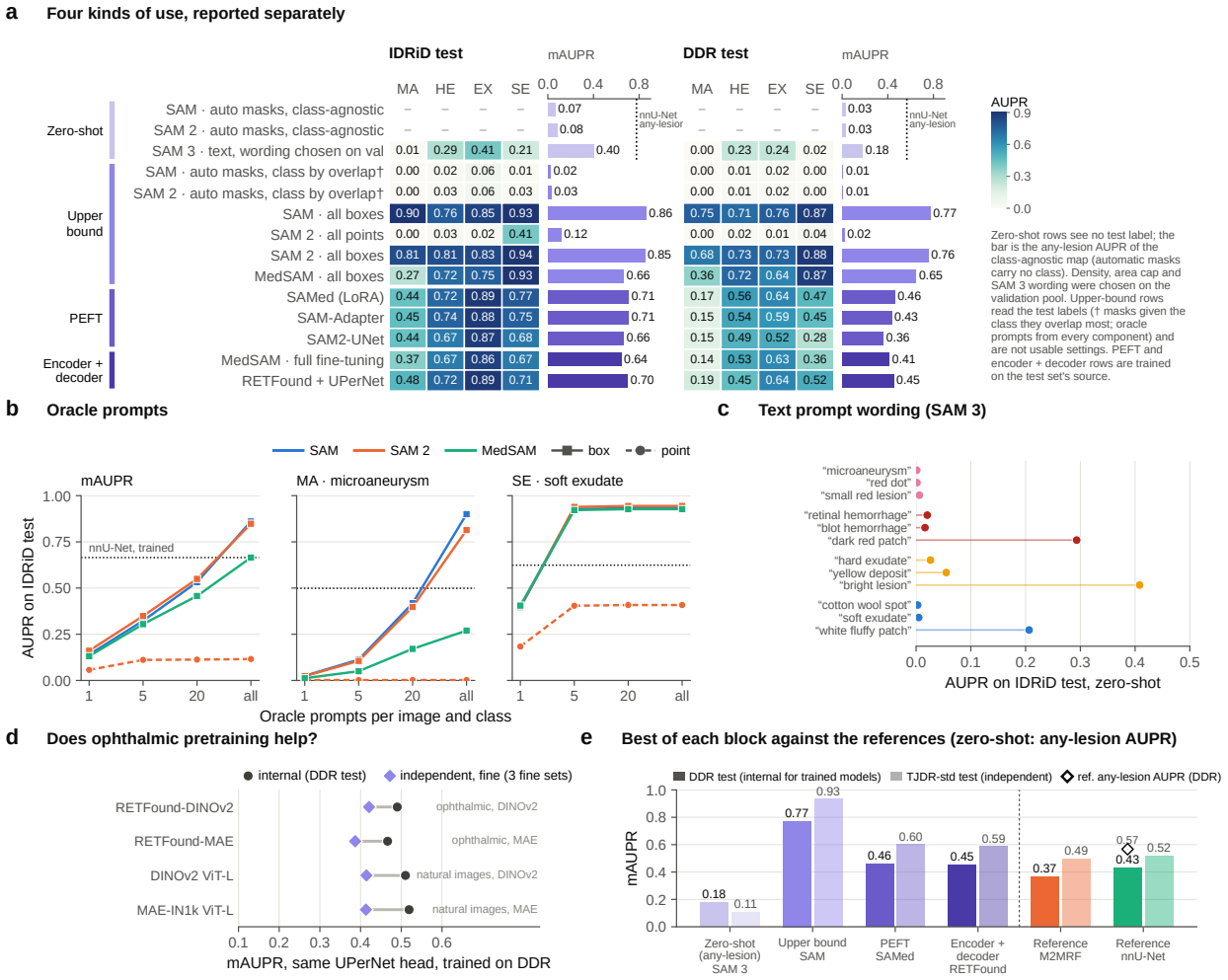


Fig. 5 Foundation models under four kinds of use. a, Per-class AUPR and mAUPR on the IDRiD and DDR test sets. Zero-shot rows see no test label and are summarised by the any-lesion AUPR of their class-agnostic maps (dotted line, nnU-Net on the same metric where measured); upper-bound rows read the test labels (automatic masks assigned the class they overlap most, prompts from every ground-truth component); the two trained settings are trained on the source of the test set. **b**, Oracle prompts: AUPR on IDRiD against the number of ground-truth boxes or points given per image and class; dotted line, nnU-Net trained on IDRiD. **c**, SAM 3 with three wordings per lesion class. **d**, Four ViT-L encoders under the same UPerNet head and recipe, trained on DDR (seed 0): internal test and mean of the three fine-grained independent test sets (the two TJDR sets and MAPLES-DR); lines join each retinal encoder to the natural-image encoder trained with the same objective. **e**, Best model of each block against the fundus-specific and self-configuring references; the zero-shot bar is an any-lesion AUPR, and diamonds give the references' any-lesion AUPR on the DDR test.

Table 5 Foundation models under four kinds of use. The kinds are reported in separate blocks and are not ranked against each other: zero-shot use sees no test label (its density, area cap and wording are chosen on the validation pool); the upper-bound block reads the test labels (masks classed by their overlap with the ground truth, prompts derived from every ground-truth component) and is not a usable setting; the two trained settings follow the unified protocol. AUPR on the IDRiD and DDR test sets. **Bold** and underlined mark the best and second-best rows of each column within a block.

			IDRiD test					DDR test				
Model	**Use**	**Backbone**	MA	HE	EX	SE	**mAUPR**	MA	HE	EX	SE	**mAUPR**
Zero-shot: no test label; density, area cap and wording chosen on the validation pool (summary = any-lesion AUPR‡)												
SAM	automatic masks, class-agnostic	ViT-H	–	–	–	–	.07	–	–	–	–	.03
SAM 2	automatic masks, class-agnostic	Hiera-L	–	–	–	–	.08	–	–	–	–	.03
SAM 3	text, wording chosen on validation	–	.01	.29	.41	.21	**.40**	.00	.23	.24	.02	**.18**
Upper bounds that read the test labels (not a usable setting)												
SAM	automatic masks, class by overlap†	ViT-H	.00	.02	.06	.01	.02	.00	.01	.02	.00	.01
SAM 2	automatic masks, class by overlap†	Hiera-L	.00	.03	.06	.03	.03	.00	.01	.02	.00	.01
SAM	oracle boxes, every component	ViT-B	**.90**	.76	**.85**	.93	**.86**	**.75**	.71	**.76**	.87	**.77**
SAM 2	oracle points, every component	Hiera-L	.00	.03	.02	.41	.12	.00	.02	.01	.04	.02
SAM 2	oracle boxes, every component	Hiera-L	.81	**.81**	.83	**.94**	.85	.68	**.73**	.73	**.88**	.76
MedSAM	oracle boxes, every component	ViT-B	.27	.72	.75	.93	.66	.36	.72	.64	.87	.65
Parameter-efficient fine-tuning (trained on the source of the test set)												
SAMed	LoRA	ViT-B	.44	.72	**.89**	**.77**	**.71**	**.17**	**.56**	**.64**	**.47**	**.46**
SAM-Adapter	adapters	ViT-H	**.45**	**.74**	.88	.75	.71	.15	.54	.59	.45	.43
SAM2-UNet	adapters + U-shaped decoder	Hiera-L	.44	.67	.87	.68	.66	.15	.49	.52	.28	.36
Encoder + decoder (trained on the source of the test set)												
MedSAM	full fine-tuning	ViT-B	.37	.67	.86	.67	.64	.14	**.53**	.63	.36	.41
RETFound	MAE encoder + UPerNet	ViT-L	**.48**	**.72**	**.89**	**.71**	**.70**	**.19**	.45	**.64**	**.52**	**.45**
M2MRF	reference, Table 4	–	.45	.63	.87	.68	.66	.15	.41	.53	.39	.37
nnU-Net	reference, Table 4	–	.50	.67	.87	.62	.66	.16	.50	.61	.45	.43

‡The zero-shot block is summarised by the any-lesion AUPR (union of the four classes against the pixelwise maximum of the maps); automatic masks carry no class, so their per-class cells are empty, and SAM 3's per-class cells are its own text prompts. The any-lesion AUPR of the references on the same test sets (IDRiD / DDR): M2MRF – / –; nnU-Net 0.78 / 0.57 (– where the run predates the metric).

†Each automatic mask is assigned the class it overlaps most in the ground truth, an upper bound for class-agnostic proposals. Scores of automatic masks use the predicted IoU.

Table 6 Unified retraining: independent tests. The checkpoints of Table 4 are applied without any adaptation to six test sets from sources never used for training. Values are mAUPR averaged over the seeds of Table 4; the DiaRetDB1 column is the official image-level ROC AUC on DiaRetDB1, averaged over the four lesion types, because its region-consensus ground truth biases pixel precision. Δ is the mean of the three fine-grained independent test sets (TJDR standard-field, TJDR ultra-widefield and MAPLES-DR) minus the internal test. Average rank is over the eight test sets of Fig. 4e (Friedman χ^2 = 97.9; Nemenyi critical difference 9.90). Per-class values with 95% bootstrap intervals are in Supplementary Table 4.

	Trained on IDRiD								Trained on DDR								
		TJDR								TJDR							
Model	Int.	Std	UWF	Maples	FGADR	Ret.-L.	DiaRet	Δ	Int.	Std	UWF	Maples	FGADR	Ret.-L.	DiaRet	Δ	**Avg. rank**
General segmentation																	
U-Net	.64	.44	.20	.34	.15	.15	.85	−.32	.37	.49	.17	.35	.16	.15	.84	−.03	12.38
DeepLabv3	.54	.43	.27	.29	.23	**.26**	.87	−.21	.38	.50	.13	.34	.25	**.28**	.90	−.05	9.25
HRNet-W48	.70	.45	.25	.40	.25	.19	.86	−.33	.45	**.62**	.17	.45	.27	.25	.87	−.03	4.13
Swin-Unet	.69	.47	.22	.40	.20	.22	**.91**	−.33	.43	.56	.18	.46	.24	.26	.92	−.03	5.25
FCT	.30	.01	.03	.14	.07	.04	.83	−.24	.18	.18	.05	.14	.07	.07	.82	−.05	17.88
Self-configuring																	
nnU-Net	.66	.43	.20	.40	.24	.19	.83	−.32	.43	.52	.20	.44	.26	.21	.82	−.04	7.38
nnU-Net ResEnc-M	.63	.42	.19	.36	.20	.14	.82	−.31	.43	.57	.17	.47	.27	.24	**.94**	−.03	9.00
Fundus-specific																	
M2MRF	.66	.40	.15	.31	.08	.11	.73	−.37	.37	.49	.18	.32	.21	.15	.78	−.03	14.63
HACDR-Net	.54	.49	.16	.29	.06	.10	.82	−.22	.25	.35	.20	.26	.10	.11	.74	+.02	16.75
HRDecoder	.65	.39	.19	.38	.20	.13	.80	−.33	.30	.50	.20	.30	.19	.13	.82	+.04	13.63
PMCNet	.65	.43	.26	.34	.21	.18	.81	−.31	.31	.46	.20	.39	.21	.19	.80	+.04	11.63
WSRFNet	.58	.35	.20	.31	.20	.13	.72	−.30	.23	.33	.14	.31	.07	.07	.81	+.02	17.38
CLC-Net	.66	.46	.21	.39	.17	.19	.79	−.31	.41	.50	.13	.35	.18	.15	.81	−.09	11.75
TC-Net	.65	.48	.21	**.41**	.21	.18	.88	−.28	.36	.47	.19	.36	.21	.18	.86	−.02	9.75
Foundation models, adapted																	
SAMed	**.71**	**.53**	.25	.40	**.28**	.24	.88	−.31	**.46**	.60	.21	**.47**	**.29**	.27	.89	−.03	**1.63**
SAM-Adapter	.71	.44	.21	.36	.22	.19	.71	−.37	.43	.54	.20	.37	.25	.19	.76	−.06	8.50
SAM2-UNet	.66	.53	.23	.37	.25	.19	.79	−.29	.36	.51	.20	.40	.25	.23	.84	+.01	7.50
MedSAM (full FT)	.64	.45	**.29**	.37	.26	.21	.84	−.27	.41	.55	**.24**	.42	.27	.26	.90	−.01	6.13
RETFound + UPerNet	.70	.45	.25	.40	.23[e]	.21[e]	.85	−.33	.45	.59	.11	.46	.27[e]	.27[e]	.88	−.06	5.50
Ensembles of the rows above (pixelwise mean of the probability maps; members chosen on validation data)																	
E-top3	.74	.56	.28	.44	.27	.25	.88	−.32	.51	.66	.23	.50	.30	.29	.89	−.05	–
E-top5	.75	.60	.29	.45	.29	.25	.88	−.31	.51	.65	.24	.51	.30	.29	.89	−.04	–
E-1perfam	.74	.56	.28	.45	.26	.24	.87	−.31	.49	.63	.23	.49	.28	.28	.89	−.05	–
E-all	.76	.59	.31	.46	.28[p]	.22[p]	.91	−.31	.50	.64	.27	.50	.29[p]	.28[p]	.87	−.04	–
E-fam. general	.73	.51	.28	.41	.24	.19	.88	−.33	.48	.61	.23	.47	.27	.27	.87	−.05	–
E-fam. auto	.69	.48	.23	.41	.24	.18	.81	−.32	.46	.57	.21	.47	.27	.23	.88	−.04	–
E-fam. fundus	.74	.57	.29	.44	.25[p]	.21[p]	.92	−.30	.44	.57	.25	.45	.25[p]	.22[p]	.85	−.02	–
E-fam. foundation	.76	.61	.30	.44	.29	.25	.86	−.31	.51	.66	.25	.49	.30	.29	.89	−.04	–
Human reference: one DiaRetDB1 expert against the consensus of the other three (leave-one-expert-out; no training source, same value under both blocks)																	
Expert 01	–	–	–	–	–	–	.87	–	–	–	–	–	–	–	.87	–	–
Expert 02	–	–	–	–	–	–	.93	–	–	–	–	–	–	–	.93	–	–
Expert 03	–	–	–	–	–	–	.93	–	–	–	–	–	–	–	.93	–	–
Expert 04	–	–	–	–	–	–	.91	–	–	–	–	–	–	–	.91	–	–
Expert range	–	–	–	–	–	–	.87–.93	–	–	–	–	–	–	–	.87–.93	–	–

Granularity of the test set: ■ fine ■ mixed ■ coarse ■ region consensus. Int., internal test (official test set of the training source). Std, standard field (35–50°); UWF, ultra-widefield (133°), predicted at native resolution with sliding windows; Maples, MAPLES-DR (official test split, 60 images). Retinal-Lesions excludes its grey-level 127 regions. Best and second-best are marked among the measured rows of a column. [e], not fully independent: RETFound's colour-fundus encoder was pretrained (without labels) on a corpus that contains the Kaggle EyePACS images, from which Retinal-Lesions and part of FGADR are drawn (Methods; average rank without these two test sets in the text). [p], ensemble computed without a registered member whose maps could not be produced on that set (named in the run's record).

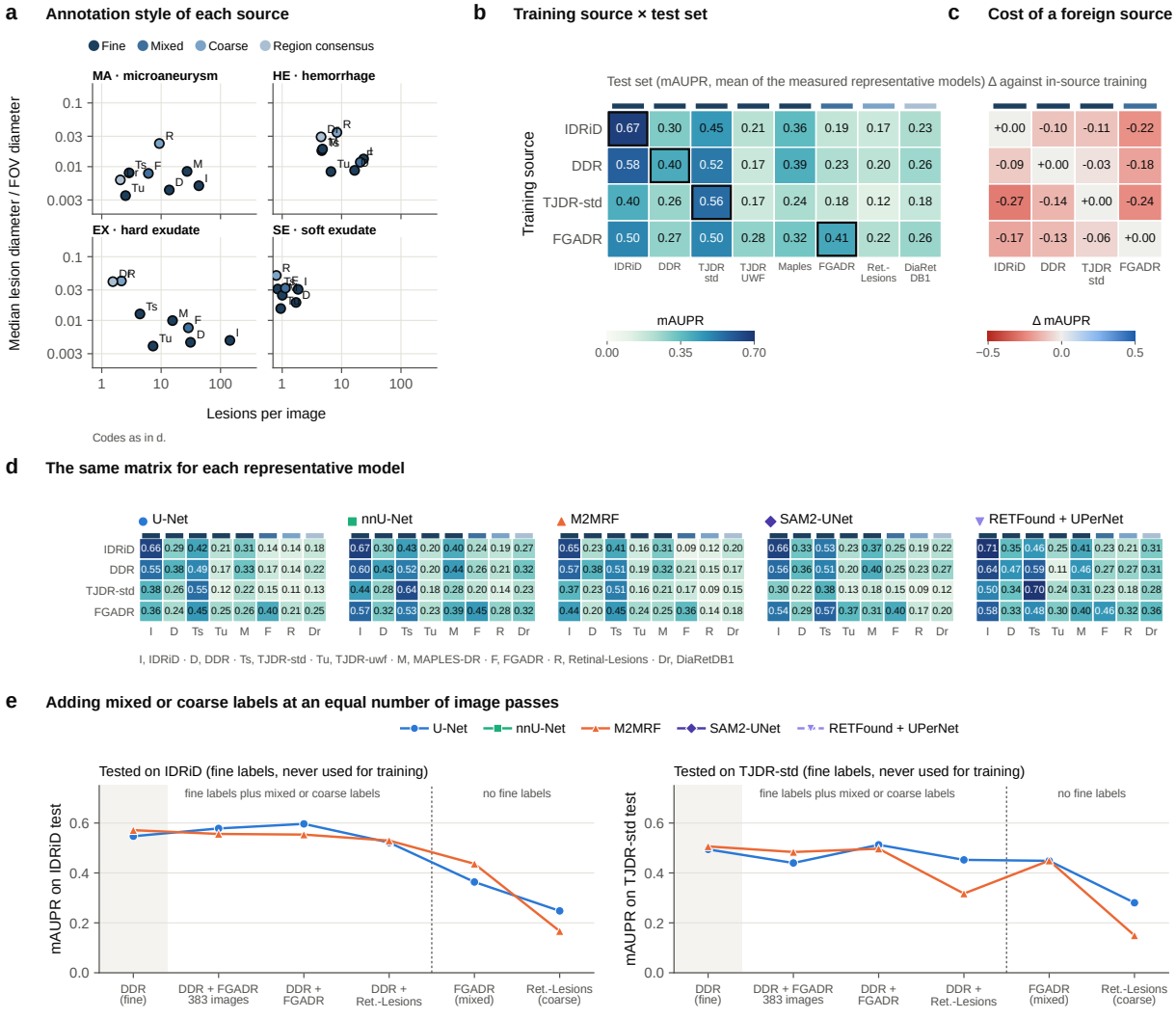


Fig. 6 Effects of annotation granularity and training source. a, Annotation style: connected components per image against their median diameter relative to the field-of-view width, per lesion class and source, over all images of each source (Methods). **b**, mAUPR for each training source (rows) and test set (columns), mean of five representative models, seed 0; outlined cells are internal tests. **c**, Difference to training on the source of the test set. **d**, The matrix of **b** for each model. **e**, Training sets of different granularity at an equal number of image passes, tested on two fine-grained test sets (U-Net and M2MRF, seed 0).

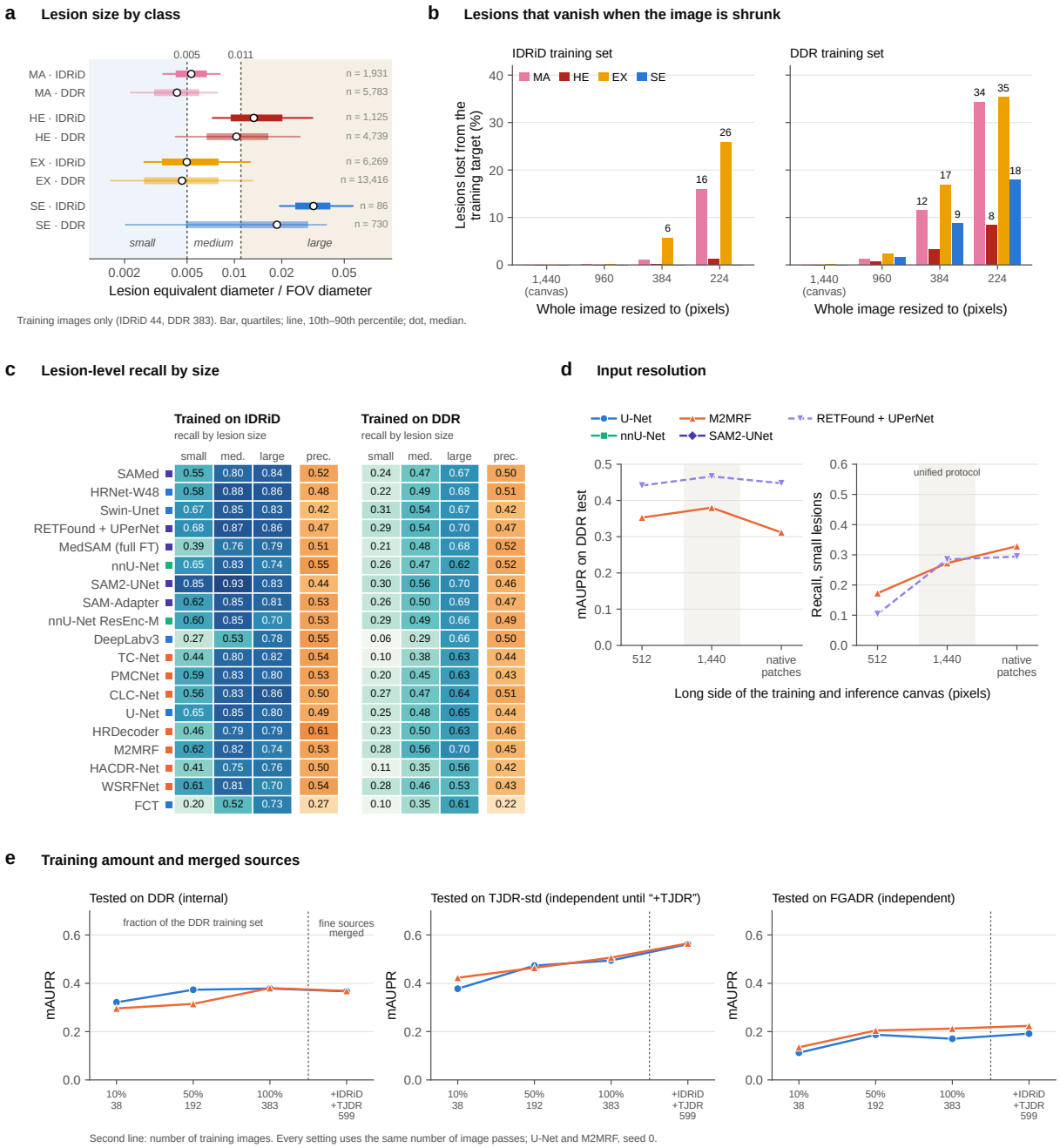


Fig. 7 Effects of lesion size, input resolution and training amount. a, Lesion diameter relative to the field-of-view diameter on the training images; dashed lines, the frozen boundaries between small, medium and large lesions. **b**, Share of connected components that vanish from the training target when the whole image is resized to the given size. **c**, Lesion-level recall by lesion size and lesion-level precision on the internal tests, at the threshold chosen on the validation set. **d**, mAUPR and recall of small lesions on DDR against the long side of the canvas (M2MRF and RETFound with UPerNet, seed 0). **e**, mAUPR against the fraction of the DDR training set and after the fine-grained sources are merged in (U-Net and M2MRF, seed 0); the number of image passes is the same in every setting.

Table 7 One factor at a time. One or two representative models are retrained with a single factor changed and everything else fixed to the unified protocol (seed 0). Each cell is the value followed by the change against the reference row of its block; – marks a model the block does not use. The last column averages the models of the block; its shading follows the change. Differences smaller than twice the median seed standard deviation of the main table are not interpreted.

Setting	U-Net	M2MRF	RETFound+UPer.	Mean
Annotation granularity of the training set: mAUPR on IDRiD test, equal image passes; U-Net and M2MRF				
DDR ■ (ref.)	.55	.57	–	.56
DDR + FGADR, 383 images ■■	.58 ▲ .03	.56 ▼ .02	–	.57 +.01
DDR + FGADR ■■	.60 ▲ .05	.55 ▼ .02	–	.58 +.02
DDR + Retinal-Lesions ■■	.52 ▼ .03	.53 ▼ .04	–	.53 −.03
FGADR ■	.36 ▼ .18	.44 ▼ .13	–	.40 −.16
Retinal-Lesions ■	.25 ▼ .30	.17 ▼ .40	–	.21 −.35
Input resolution, long side in pixels: mAUPR on DDR test; M2MRF and RETFound + UPerNet				
512	–	.35 ▼ .03	.44 ▼ .03	.40 −.03
1,440 (ref.)	–	.38	.47	.42
native-resolution patches	–	.31 ▼ .07	.45 ▼ .02	.38 −.04
Input resolution: lesion-level recall of small lesions on DDR test				
512	–	.17 ▼ .10	.10 ▼ .18	.14 −.14
1,440 (ref.)	–	.27	.28	.28
native-resolution patches	–	.33 ▲ .06	.29 ▲ .01	.31 +.03
Training amount: mAUPR on DDR test; U-Net and M2MRF				
DDR, 10% (38 images)	.32 ▼ .06	.30 ▼ .08	–	.31 −.07
DDR, 50% (192)	.37 ▼ .01	.31 ▼ .07	–	.34 −.04
DDR, 100% (383) (ref.)	.38	.38	–	.38
+ IDRiD + TJDR standard field (599)	.37 ▼ .01	.37 ▼ .01	–	.37 −.01
Loss, architecture fixed: mAUPR on DDR test; U-Net				
cross-entropy (ref.)	.36	–	–	.36
batch Dice	.24 ▼ .12	–	–	.24 −.12
cross-entropy + batch Dice (unified protocol)	.38 ▲ .02	–	–	.38 +.02
cross-entropy + per-image Dice (registered recipe)	.12 ▼ .24	–	–	.12 −.24

Granularity of the labels used for training: ■ fine ■ mixed ■ coarse. In the training-amount block the number of image passes is the same in every row. The FGADR row of the granularity block is the FGADR-trained model of the cross-source experiment.

Discussion

Published performance rankings are not a sufficient basis for choosing a method for diabetic retinopathy lesion segmentation. A published score combines three properties that behave differently: whether the number can be reproduced, how the model performs when training and evaluation are fixed, and whether its performance carries over to data from other sources. Each property is shaped by factors outside the architecture: the evaluation implementation, the annotation conventions of the data and the imaging conditions. Separating the three properties is what makes a comparison informative.

Numerical reproducibility. A reported score describes one checkpoint under one evaluation. The evaluation is often unstated, and the most widely reused implementation adds a model-dependent amount to the AUPR, enough to reorder models. Baselines are mostly carried from earlier papers, so a comparison table places a new method beside numbers produced under other settings. Released training code did not reproduce the released weights. A reported number therefore cannot be read as a property of the method; it becomes comparable only when the implementation is stated and the baselines are rerun under the same evaluation.

Performance under controlled conditions. With data, training amount and evaluation fixed, the differences among the leading models were of the size of their variation between seeds, and most models could not be separated from the first. The published ranking did not predict this order. Adapted foundation models and a general high-resolution network were at the top, in line with reassessments in three-dimensional medical segmentation[27], and lesion-specific decoders did not improve on a general head with the same backbone. The large departures came from identifiable recipe components, such as a coarse output stride or a per-image Dice term, rather than from a family of methods. For foundation models, the mode of use decided the result. Without training they did not find lesions, although they delineated a lesion well once it was located. Once trained on the task they were competitive. Their advantage appears to come from the encoder and the scale of its pretraining, to which retinal pretraining added nothing in our comparison.

Generalization across data sources. Performance followed the data more than the model. Models kept their performance on sources labelled in the same fine-grained style and lost a third to a half of it on coarse, region-level or ultra-widefield data. Internal rank predicted independent rank well on the first kind of source, less well on coarse and region-labelled data, and not at all on ultra-widefield images. A source labelled like the test data was worth more than a larger source labelled differently, as Playout and Cheriet observed for U-Net-type baselines[25], and adding coarse labels to fine ones did not help. Small lesions, which carry the earliest signs of disease, were the most sensitive to input resolution. These effects were as large as most differences between architectures in the main table. An internal test thus describes performance on images labelled and acquired like the training set, a conclusion shared with assessments in other fields[28].

Implications. Method papers should report the three properties separately. For reproducibility, they should state the split and the AUPR implementation, compute the exact AUPR inside the field of view at the original resolution, rerun every baseline, and release probability maps so that any metric can be recomputed. For controlled comparisons, they should include a self-configuring baseline and report several seeds and intervals over test images. For generalization, they should report at least one test set from a source not used for training, without averaging across annotation conventions. Users choosing a model should weigh the training data as much as the architecture: fine-grained labels from an imaging setting like their own, and the recall of small lesions on an independent source, are more informative than a mean AUPR on IDRiD. Where computation allows, an ensemble of several retrained models is the strongest option in our results, and the reference model offers a smaller alternative.

Limitations and outlook. Most comparisons rest on two training sources. Nine models of the main table have three seeds, and the factor experiments use one to five representative models with one seed, so we interpret only differences larger than the seed variation of the main table. IDRiD has 27 test images, and the ultra-widefield set comes from a single centre and device. RETFound was pretrained on the EyePACS images from which Retinal-Lesions and part of FGADR are drawn. Pixel-level labels are graders' judgements, and the clinical consequences of segmentation errors, such as whether a missed or spurious lesion changes the grade or the referral, are not assessed here. We are reviewing model errors with ophthalmologists to address them; in screening, the value of an automated system also depends on image quality, workflow and how graders use its output[62–64]. Future rounds will add multi-seed factor experiments, a ranked row for natural-image encoders under the same head, the tier-2 models and further test sets (e-ophtha[65], ROC[66] and ultra-widefield OCTA).

Methods

Workflow of model evaluation

The design is summarised in Fig. 1a and in the first section of the Results; the sections below give the details in the same order. Each experiment is named in the text by what it does; the identifiers under which the runs, the results table and the planning documents of the repository file them are listed in Supplementary Table 18. The evaluation protocol and all splits were committed to the repository before the first main-table run. Before that, pipeline checks scored a reference U-Net and the released M2MRF checkpoints on the IDRiD and DDR test sets, and the field-of-view rule and the lesion-size boundaries were revised once from training images. The training amount and the selection rule were set from training and validation curves. The only recipe changed after a model had been scored is the U-Net loss (Rare-class collapse, below).

Data collection

Training sources. IDRiD[13] provides 81 images of 4288×2848 pixels with pixel masks for MA, HE, EX and SE; the official split is 54 training and 27 test images. A missing mask file means that the image has no lesion of that class. DDR[15] provides 757 images with the official split 383 / 149 / 225.

Independent test sources. TJDR[18] contains 561 images from one hospital: 257 standard-field images (2048×2048, 35–50°) and 304 ultra-widefield images (3912×3912, 133°), annotated with polygons by two ophthalmologists independently and adjudicated by a third; the four classes are mutually exclusive. The two fields are treated as two datasets. MAPLES-DR[19] adds pixel labels for 198 MESSIDOR[67] images (45°; 1440×960 to 2304×1536 pixels): seven retinologists drew MA, HE, EX and SE at 1500×1500 on the field-of-view crop, MA, HE and EX starting from a network's pre-segmentation and SE from scratch; the labels are placed back on the original image, and only the official test split of 60 images is used. The FGADR Seg-set[16] contains 1,842 images of 1280×1280 pixels with masks for the four classes and, as extension labels, intraretinal microvascular abnormalities (159 images) and neovascularization (49); the images with the file-name suffix _3 appear to come from Kaggle EyePACS (90 of them are the same photographs as Retinal-Lesions images, and no image with another suffix is). Retinal-Lesions[17] (release 20191227) contains 1,593 images from EyePACS[68], cropped to the field of view and resized to 896×896, from 1,505 patients, annotated with ellipses and polygons by at least three of 45 graders. DiaRetDB1 v2.1[23] contains 89 images of 1500×1152 pixels (50°) in which four experts independently marked regions (4,135 circles, 763 ellipses, 1,036 polygons), each with a confidence level.

Granularity tiers. Sources are assigned to four tiers: fine (IDRiD, DDR, TJDR standard field, TJDR ultra-widefield, MAPLES-DR), mixed (FGADR), coarse (Retinal-Lesions) and region consensus (DiaRetDB1).

Label mapping. Only the four classes are mapped; other classes are stored as extension labels and do not enter the four-class mean. TJDR: palette 1, 2, 3, 4 to EX, HE, MA, SE. FGADR: HardExudate, Hemohedge, Microaneurysms, SoftExudate. Retinal-Lesions: hard_exudate, retinal_hemorrhage, microaneurysm, cotton_wool_spots; vitreous and preretinal hemorrhage are not merged into HE. DiaRetDB1: Hard_exudates, Haemorrhages, Red_small_dots, Soft_exudates. MAPLES-DR: Microaneurysms, Hemorrhages, Exudates, CottonWoolSpots; its uncertain red lesions (MA or HE) and uncertain bright lesions (EX, SE or drusen) are ignored for the two classes they could belong to, and drusen and neovascularization are extension labels.

Model collection

The literature search was frozen at 30 September 2026. A fixed query (three concepts in title or abstract: diabetic retinopathy; lesion, microaneurysm, exudate, hemorrhage or cotton wool; segmentation) was run on PubMed, Europe PMC and OpenAlex for 2015 onwards. It identified 1,389 records, of which 948 passed the Boolean gate and 764 remained after de-duplication by title; 112 entered the full-text queue. Screening of a priority list (27 DOIs, one dataset lead and one further paper found when a mis-assigned DOI was corrected) required colour fundus photographs, pixel-level segmentation of at least two of the four classes and an obtainable public dataset; it included 24 methods and 2 datasets and excluded 3 records. Together with a snowballed corpus the ledger holds 86 records. Only the authors' own repositories count as official code.

39 model variants were registered in three tiers (22, 6 and 11). Each has a reproducibility level: R0, no code; R1, model definition only; R2, inference code and weights; R3, training code; R4, training code, weights, splits and environment. Each is assigned to evaluation tracks: A, author weights; B, unified retraining of the official code (for U-Net, whose authors released a Caffe package, a reference PyTorch implementation); C, re-implementation from the paper, marked in the tables. A re-implementation fixes every unstated choice in a written list of assumptions before any experiment, and its parameter count is compared with the published one. Methods whose repository holds the model but no training loop (WSRFNet) or whose code cannot run on current hardware (PMCNet, TensorFlow 1.13) are retrained under Track B with the loss and optimizer the paper states, written into the wrapper and marked in the tables. HRDecoder, which has official code and weights, was added after a targeted check for code-available methods that the frozen query had not returned. Authors of methods without code received a fixed e-mail on 3 October 2026 asking for code, weights and splits, with a deadline of four weeks; where code exists without weights, the weights are requested separately. Non-responses and failures remain in the denominator of the reproducibility funnel.

Preprocessing

Unified layout. Every source is exported to the same layout: the image at original resolution with the bytes of the release, one binary mask per class, a field-of-view mask and the split lists. A missing mask file is an all-negative image for that class; the image stays in the split.

Field of view. The field-of-view (FOV) mask is the largest connected component of pixels with luminance $0.299R + 0.587G + 0.114B > 10$, with holes filled. A fixed threshold is used rather than Otsu's, which removed the darker half of the retina in several training images. With the fixed threshold, the largest share of lesion pixels lying more than 2 pixels outside the mask on the training images is 2.3×10^{-5} (IDRiD hemorrhages) and 0 on DDR. For the independent sources one rule is added: if the four 40×40 corner blocks are uniform (1st to 99th percentile within 5 grey levels), the threshold is the larger of 10 and the corner median plus 3. It changes one of 4,085 images, a DiaRetDB1 image with a uniform grey background.

Canvas. For training and inference the image is cropped to the FOV and resized so that its long side is 1,440 pixels. Predictions are mapped back to the original resolution by bilinear interpolation, set to zero outside the

FOV and stored as 16-bit images (probability = value / 65535). All metrics are computed offline from these maps.

Source-specific handling. FGADR masks have anti-aliased edges with values 1–254; a pixel is positive at ≥ 128, and the number of pixels dropped is recorded per class. In Retinal-Lesions, value 255 is lesion and value 127, which the authors use for annotations that do not follow the grading guideline, is written to a per-class ignore region that is skipped by all metrics and, when this source is used for training, by the loss. In DiaRetDB1, circles, ellipses and polygons are rasterized at pixel centres; for one expert and class the highest confidence per pixel is kept, with High = 1.0, Medium = 0.75 and Low = 0.25; the four experts are averaged and a pixel is positive at ≥ 0.75, the threshold of the original protocol. With this rule six of the eight positive-image counts of the original paper are matched exactly and two differ by one image (the preparation report in the code repository); the union of all expert marks is kept as a lenient ground truth for sensitivity analysis.

Exclusive training targets. Models with a softmax head need one label per pixel. Overlapping pixels are written in the order EX, HE, SE, MA, later classes overwriting earlier ones. This removes less than 0.5% of the positive pixels of any class in the training sets (at most 0.34% at original resolution). Evaluation always uses the four original, possibly overlapping masks.

Construction of training, validation and test sets

Official splits are never changed. New splits use the generator PCG64 with seed 20261003, and the lists were committed before any training on the source concerned. Where a split is stratified, quotas per stratum follow the largest-remainder method. IDRiD: 10 of the 54 official training images, drawn at random from the sorted identifiers, are held out for validation (44 / 10 / 27); author-recipe retraining uses all 54. DDR: official 383 / 149 / 225. TJDR: the official test set is kept and 15% of the official training images of each field are drawn for validation, giving 172 / 30 / 55 (standard) and 209 / 37 / 58 (ultra-widefield). FGADR: stratified by file-name suffix and DR grade, 30% test and 15% of the remainder for validation (1,096 / 193 / 553). Retinal-Lesions: split by patient, stratified by the higher grade of the two eyes (951 / 164 / 478). DiaRetDB1: official 28 / 61; the 28 training images are used only to choose thresholds.

Test scenarios and leakage audit

The internal test is the official test set of the training source. Independent tests come from sources excluded as a whole from training and from internal testing, as in the assessment we follow[28]. Every main-table checkpoint is applied to the test splits of the five independent sources. Ultra-widefield images are predicted at their native resolution with sliding windows, because matching pixels per degree to the 1,440-pixel canvas would require upsampling; a version resized to 1,440 pixels is reported for comparison. The per-degree scale is defined by the field of view of the training source; for a run whose training sources do not fix one (the factor cells trained on the standard-field TJDR images, on FGADR, or on merged sources), only the 1,440-pixel version is run and stands for the ultra-widefield result. For author weights, a source that the original model was trained on is removed from its independent tests. Near-duplicate images are searched with one signature over all pairs of images of the prepared sources, within and between sources: the green channel of the field-of-view box, resized to 192×192 and filtered with a 5×5 black top-hat so that only thin dark structures (vessels and red lesions) remain, compared by normalised cross-correlation inside the field of view (the larger of the direct and the mirrored comparison). Photographs of the same eye score 0.5–1.0 and photographs of different eyes generally below 0.4, and every pair above 0.4 was checked visually. A test image whose correlation with a training or validation image of a source reaches 0.5 (0.33 within FGADR) is left out of that source's independent tests and listed (Supplementary Table 15). IDRiD and DDR share no

eye with any other source, so no main-table checkpoint loses a test image; the exclusions apply only when FGADR, Retinal-Lesions or TJDR images are used for training (factor cells). The pretraining corpora of the foundation models cannot be searched image by image, and their overlap with the test sets is determined from the sources. RETFound's colour-fundus encoder[50] was pretrained without labels on 904,170 images, of which 88,702 are the complete Kaggle EyePACS set; Retinal-Lesions consists of EyePACS images (its file names are EyePACS identifiers), and so do the FGADR images with the suffix _3. RETFound's results on Retinal-Lesions and FGADR are therefore not fully independent and are marked in Table 6. The DINOv2 version of RETFound in the encoder comparison does not list EyePACS among its pretraining data, MedSAM saw fundus images in training only for optic disc and cup segmentation, and the web-scale image corpora behind SAM, SAM 2, SAM 3 and DINOv2 cannot be audited image by image.

Evaluation of the original models

Released checkpoints (author weights). Released checkpoints are run in a current framework; for M2MRF, whose repository requires a framework version that does not support the cards used here, the forward pass was checked by reproducing the repository numbers with the authors' function. Each checkpoint is reported with three numbers: the paper, the authors' evaluation function, and this study's protocol. The authors' function for M2MRF evaluates the whole image, overwrites overlapping labels in the order EX, HE, SE, MA and computes AUPR from 11 thresholds.

Author-recipe reruns. M2MRF and HRDecoder are retrained with the authors' pipelines, augmentation and training set (all 54 IDRiD training images; three M2MRF seeds and one HRDecoder run, plus a first M2MRF run at an effective batch of 1 that is reported separately) and scored at the last iteration without checkpoint selection, with the authors' function and with this study's protocol. For M2MRF: binary Dice loss (smoothing 10^{-5}); SGD with learning rate 0.01, momentum 0.9 and weight decay 5×10^{-4}; polynomial decay with power 0.9; 40,000 iterations on IDRiD and 60,000 on DDR; random crops of 960×1440 after resizing; photometric distortion; HRNet-W48 initialized from ImageNet. For HRDecoder: the authors' segmentor and loss, images resized to 2880×1920 with scale jitter 0.5–2, random 1920×1920 crops, flips and rotations of $\pm 45^{\circ}$, SGD 0.01 with 1,500 warm-up steps and linear decay over 20,000 steps, four images per step; their four-process synchronized batch normalization becomes one process with four images. The authors' runner evaluates on the test split every 2,000 iterations and keeps the best checkpoint; the rerun reports the last step, and what that test-split selection would have returned is given separately.

Unified retraining

Shared settings. Data, splits, the 1,440-pixel canvas, augmentation, the training amount in pixels, checkpoint selection, inference geometry and evaluation are the same for every model. Each model keeps its loss, optimizer, learning rate, weight decay, schedule shape and effective batch from its source recipe (Table 8); the schedule is stretched over the shared number of steps.

Training amount. One equivalent epoch draws as many patch pixels as the canvases of the training split contain, so that a model trained on 224×224 patches draws about 18 times more patches per epoch than one trained on 960×960 patches:

$$\text{steps per epoch} = \left\lceil \frac{1}{B} \left\lceil \frac{\sum_i H_i W_i}{h\,w} \right\rceil \right\rceil , \tag{1}$$

where $H_i \times W_i$ is the canvas of training image i, $h \times w$ the patch and B the effective batch. The number of equivalent epochs was fixed per dataset by a budget probe: U-Net and M2MRF were trained once (seed 0) at twice a provisional amount, and only training loss and validation mAUPR were read. The validation curve was smoothed with a centred three-point mean; if the smoothed maximum of any probe of a dataset lay in the

last 10% of the steps, the amount was doubled. This gave 1,000 equivalent epochs for IDRiD and 400 for DDR; no test metric was used.

Patches and inference. Models with a fixed input size (Swin-Unet and RETFound with UPerNet, 224; FCT, 384; CLC-Net, TC-Net and SAMed, 512; SAM-Adapter and MedSAM, 1,024) are trained on patches of that size cut from the canvas and predict with sliding windows of the same size at an overlap of 0.5; HRDecoder keeps its authors' windowed inference at 960. All other models are trained on 960×960 patches and predict on the whole canvas (Table 8).

Augmentation. Random scaling by 0.5–2 before cropping, horizontal flip with probability 0.5, rotation by multiples of 90°, and photometric distortion with probability 0.5 (brightness ± 32, contrast and saturation 0.5–1.5, hue $\pm 18°$). The randomness of a sample depends only on the seed and the sample index.

Checkpoint selection. A run is validated 20 times at equal intervals and the checkpoint with the highest validation mAUPR inside the FOV is scored; the last step is kept for a sensitivity analysis. A model whose best checkpoint falls in the last two validations is flagged as possibly under-trained. The test split is not read during training.

Reproducibility. A resume file stores model, optimizer, step, selection state, random-number state and data order; a run resumed after an interruption is bit-identical to an uninterrupted one. Each run writes its environment (host, card, driver, framework, seed, commit at process start, parameter count, wall time) and its recipe with every deviation.

Rare-class collapse. The registered recipe of the reference U-Net uses a binary cross-entropy plus a sigmoid Dice term computed per image and class with smoothing 1. On a patch without positives, this term is minimized by an all-zero output with a gradient of about 1 per pixel. On a patch with a few positive pixels, the gradient per pixel is about $2/(\text{number of positive pixels} + 1)$. Under patch training, 83% of DDR patches and 67% of IDRiD patches contain no soft exudate. In the budget probe, the rare-class channels collapsed to zero (soft exudates on IDRiD; microaneurysms, hemorrhages and soft exudates on DDR). A Dice term pooled over the batch, as in nnU-Net, is not affected. The U-Net's Dice term was therefore pooled over the batch per class with smoothing 10^{-5}. This revision was made after the registered U-Net run had been scored on the test sets. It was based on the training and validation curves of that run and of a diagnostic run with the pooled term (validation mAUPR on DDR 0.544 against 0.127 with the per-image term, and 0.569 on IDRiD). Both versions are reported, the registered one in Supplementary Table 6. SAM2-UNet computes a weighted IoU term per image and class with smoothing 1, and MedSAM a Dice term per image and class with squared predictions and smoothing 10^{-5}. For both, a rule was registered before training: if the validation AUPR of a rare class stays at its prior level throughout training, the term is pooled over the batch and the model is retrained from its initial weights. The rule applied to both on DDR, where the microaneurysm and soft-exudate validation AUPR stayed at or near their prior levels while the other classes trained. The main tables show the retrained runs, and the collapsed runs are kept as the registered recipes in the supplementary tables.

nnU-Net. nnU-Net is run through its own entry point in 2D at original resolution, with its own schedule and checkpoint selection, with the same splits written into its split file and exclusive labels composed in the same overwrite order; softmax probabilities are exported, mapped to the original image and scored with the same evaluation. The default configuration and the residual-encoder preset M are both run. It uses its own preprocessing, which is marked in the tables.

Foundation models

Four kinds of use are reported separately. (a) Zero-shot: automatic masks of SAM and SAM 2, scored once as class-agnostic "any lesion" proposals and once with each mask assigned the class it overlaps most; text prompts for SAM 3 with three wordings per class. (b) Oracle prompts: boxes or points generated from ground-truth connected components, with 1, 5, 10, 20 or all components per image and class; MedSAM, which accepts boxes only, belongs here. (c) Parameter-efficient fine-tuning: SAMed, SAM-Adapter and SAM2-UNet, trained on patches cut from the canvas (512, 1,024 and 960 pixels; Table 8). (d) Encoder plus decoder: MedSAM fully fine-tuned, and RETFound with a UPerNet head. To isolate the effect of ophthalmic pretraining, two pairs of ViT-L encoders fixed before any was trained are run on DDR with the same UPerNet head, recipe and training amount: RETFound's masked-autoencoder encoder against the ImageNet masked autoencoder it was initialised from, and RETFound's DINOv2 encoder against DINOv2 itself (one seed each; the pair differences are the pre-specified comparisons). The three control encoders were also trained on IDRiD under the same head and recipe, so that both pairs exist on both training sources. Settings (c) and (d) follow the unified protocol.

Evaluation of factors

Each experiment changes one factor and keeps the unified protocol otherwise; every cell is trained at the pixel budget of DDR with the DDR recipe of its model and one seed (seed 0), and we interpret only differences larger than twice the median seed standard deviation of the main table. *Training source*: the five representative models (U-Net, nnU-Net, M2MRF, SAM2-UNet and RETFound with UPerNet) are trained on the standard-field TJDR images and on FGADR and tested on all eight test sets, which with the main table gives four training sources. *Annotation granularity*: U-Net and M2MRF on six training sets at an equal number of patch pixels, namely DDR, FGADR, Retinal-Lesions, DDR with FGADR, DDR with Retinal-Lesions, and DDR with 383 FGADR images drawn once (list committed before training); the DDR and FGADR rows are the main-table and training-source runs. *Input resolution*: M2MRF and RETFound with UPerNet on DDR with a long side of 512 or 1,440 pixels (the main table) and with patches at native resolution; the HRDecoder case study of the main text trains that model at its authors' 2,880-pixel canvas and 1,920-pixel window with the equivalent epochs of the unified protocol counted at that canvas. *Training amount*: U-Net and M2MRF on 10%, 50% and 100% of the DDR training set (nested subsets drawn once) and on IDRiD, DDR and standard-field TJDR merged. *Loss*: U-Net on DDR with cross-entropy, batch Dice, their sum (the main table) and the per-image Dice of the registered recipe, with the architecture fixed. *Learning rate*: HRNet-W48 and M2MRF on DDR with the learning rate multiplied by 1/3 and 3, judged on validation mAUPR only.

Metrics

Pixel level. For each class, the positive and negative pixels of all test images inside the FOV (and outside the ignore region of that class) are pooled, and AUPR is computed exactly over the 16-bit probability levels of the stored maps; this equals the average precision

$$\mathrm{AUPR} = \sum_{k} (R_k - R_{k-1})\, P_k, \tag{2}$$

with precision P_k and recall R_k at the k-th distinct threshold. mAUPR is the mean over the four classes. Dice and intersection over union are reported at a fixed threshold of 0.5 and at the per-class threshold that maximizes Dice on the validation set. For an image without positive pixels of a class the per-image Dice is missing, not zero.

Lesion level. Lesions are 8-connected components. A ground-truth lesion is detected if any predicted positive pixel overlaps it; a predicted component is a true positive if it overlaps any ground-truth pixel of the class.

Precision, recall and F1 are pooled over the split. Lesion size is the equivalent diameter of the ground-truth component divided by the equivalent diameter of the FOV; the boundaries between small, medium and large, 0.005 and 0.011, are the medians of microaneurysms and hemorrhages on the 44 IDRiD and 383 DDR training images, rounded up to 0.001. Validation and test images were not used to set them.

Annotation style. The labelling style of each source (Fig. 6a) is measured on the prepared masks of all its images, every image once whatever its split: per class, the 8-connected components are counted and their equivalent diameter is divided by the long side of the FOV bounding box (which for a circular field equals the FOV diameter); ignore regions are removed before counting, and DiaRetDB1 is measured on the ≥ 0.75 consensus the protocol evaluates. Reported are the mean number of components per image, over all images of the source, and the median relative diameter over all components.

DiaRetDB1. The primary result follows the official image-level protocol: the image score of a class is the maximum probability inside the FOV, and the area under the ROC curve and the weighted error rates are reported. Pixel-level AUPR against the consensus regions is given separately and marked as region ground truth.

Authors' implementation. The 11-threshold AUPR is reported next to the exact value for all models, to relate our scores to published numbers.

Case study of one model's geometry. HRDecoder trains its backbone at half the resolution of a crop that is itself two thirds of the authors' 2,880-pixel canvas; the unified protocol halves the canvas and keeps that ratio. Three settings separate the recipe from the budget and the canvas: the authors' recipe retrained (the author-recipe rerun), the unified protocol at twice its training amount (for HRDecoder and, as a control, HRNet-W48 on DDR), and the unified protocol on the authors' canvas with the authors' 1,920-pixel window and equivalent epochs counted at that canvas (four times the pixels per step). All three were registered before any of them was trained.

Ensembles. Probability maps of the main-table models trained on one source are averaged pixel by pixel in five compositions fixed before any ensemble was scored: all models; the three and the five best by validation mAUPR; the best model of each family; and each family on its own. The member list is the 18 main-table rows available when the compositions were registered; MedSAM is not a member. Members are chosen on validation data only; the ensemble's own validation maps set its thresholds; the test sets and the scoring are those of the single models. Ensembles are reported below the single models and are excluded from the rank tests.

Reference model. As a starting point for users of the benchmark, two models are trained once on the merged training splits of the three fine-grained sources (IDRiD, DDR and the standard-field TJDR set; 599 images) with the recipe, canvas and pixel budget of their DDR main-table run, validated on DDR, and their probability maps are averaged as in the ensembles. The two were chosen before any merged run by their DDR validation mAUPR among the main-table models (RETFound with UPerNet, 0.643; SAMed, 0.632); the pair, the budget rule and the validation source were registered before training. The reference model is scored on every test set like the ensembles, with thresholds from the DDR validation split of the averaged maps. It will be released with its weights and maps and with a command that applies it to new images. The command detects the field of view by the rule above, runs each member's sliding-window inference on the 1,440-pixel canvas, and writes the averaged probability map at the original resolution and masks at the validation thresholds.

Human reference. On DiaRetDB1 each of the four experts in turn is scored as if it were a model: the expert's own markings, with the confidence mapping above, are the probability map, and the consensus of the other three (mean confidence ≥ 0.75) is the ground truth; thresholds come from the official training list as for

the models, and the four values of each metric are reported as a band next to the DiaRetDB1 column. A map with four confidence levels gives a four-point precision–recall curve, so this band is a lower bound on what an expert's pixel-level AUPR would be with a continuous score.

Results by DR grade. For the test sets that carry a DR grade per image (DDR, FGADR, Retinal-Lesions, MAPLES-DR), every score is also computed within grades 0–1 and within grades 2–4 from the stored per-image histograms, with the thresholds of the full test; ungradable images enter neither stratum. Image-level detection summarizes the maximum probability inside the FOV against the presence of the class in the ground truth.

Label revision. Refined IDRiD[61] re-annotates the IDRiD images as one 1024×1024 grey-level label per image (one class per pixel: background, vitreous hemorrhage, retina, fovea, vessels, optic disc, hard exudates, IRMA, hemorrhages, neovascularisation, cotton-wool spots and microaneurysms) on a crop of the retinal region. For each of the 27 test images the crop is located in the original image by a coordinate-descent search over the four edges of the box, started from our FOV box, under two resampling hypotheses (the box stretched to the square, or padded to a square and resized uniformly), and accepted only when the normalised cross-correlation between the resampled original and the release reaches 0.97 (all 27: padded, 0.9988–0.9996). The official masks and the FOV are resampled through the recovered geometry with nearest-neighbour interpolation; cotton-wool spots are read as soft exudates and the other refined classes are ignored. Each IDRiD-trained checkpoint's original-resolution test maps are moved through the same geometry and scored by the protocol evaluator against both label sets, with the thresholds of its own validation run; no model is retrained or re-run. The per-class differences between the two label sets (pixels, components, intersection over union) are reported with the scores.

Statistics

Confidence intervals are 95% percentile intervals from 2,000 bootstrap replicates over test images (seed 0); each replicate pools the histograms of the drawn images and recomputes AUPR, and a replicate without positive pixels is dropped; the interval of mAUPR averages the four class values of the same replicates and drops a replicate in which any class has no positive pixel. Point estimates use the histograms at full 16-bit resolution and bootstrap replicates use per-image histograms coarsened to 4,096 bins. Two models are compared by a paired bootstrap on AUPR with shared image indices, and by a two-sided Wilcoxon signed-rank test on per-image Dice with Holm correction. Seed mean and s.d. are reported separately from bootstrap intervals. Rankings are compared across test sets with the Friedman test and the Nemenyi critical difference[69],

$$\mathrm{CD} = q_\alpha \sqrt{\frac{k(k+1)}{6N}}, \tag{3}$$

for k models and $N = 8$ test sets (two internal and six independent). Scores are the seed means of Table 4; in an independent block the score of a model is the mean over its IDRiD-trained and DDR-trained checkpoints; the DiaRetDB1 block uses the official image-level AUC on DiaRetDB1 (mean over the four classes). Agreement between published and retrained rankings is measured by Kendall's τ.

Time and resource consumption

Training and inference ran on a node with eight NVIDIA A100 cards (40 GB each), one task per card, with a queue that assigns tasks from longest to shortest expected duration; the study used about 2,000 GPU hours in total, of which about 1,650 were training under the unified protocol, about 100 the author-recipe reruns, and the rest independent tests, the inference tiers of the foundation models and the cost audit. Data export, the scoring of author weights and all metric computation ran on a workstation (RTX 4060 Ti, 16 GB). Where

a card cannot hold the effective batch of a recipe, gradients are accumulated and the deviation is recorded. Parameter count and GPU hours are recorded for every run. Inference is audited on one 1,440-pixel canvas with each model's own windows (floating-point operations, peak GPU memory and the median time of five passes on one A100) for the 15 main-table models that could be built in the audit environment (Extended Data Fig. 4 and Supplementary Table 8).

Data availability

IDRiD, DDR and TJDR are public. FGADR and Retinal-Lesions are available from their authors on request. The FGADR licence forbids redistribution of the images and of any derived data; Retinal-Lesions is distributed under application terms, and we handle it in the same way. DiaRetDB1 is public. The MAPLES-DR labels are public (CC BY 4.0); the MESSIDOR images they annotate are available from ADCIS on request. We will release all split lists, the protocol files and, for IDRiD, DDR, TJDR, MAPLES-DR and DiaRetDB1, the probability maps of every model (the maps hold no image data; DiaRetDB1 is distributed for research use). For FGADR and Retinal-Lesions we will release split lists and code only.

Code availability

The benchmark code (RetiSeg), including data export, training under the unified protocol, inference, the evaluation module, the reference model's inference entry point and the scripts that produce every figure and table of this paper, will be made publicly available. Official model repositories are referenced as pinned submodules and are not modified. The weights of the reference model's two members, with their configurations, the thresholds and a checksum list, will be released with the code; they were trained on IDRiD, DDR and TJDR, and their redistribution will follow the terms of these three sources.

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Table 8 What each model keeps under the unified protocol. Data, splits, the 1,440-pixel canvas, augmentation, the training amount in pixels, checkpoint selection and evaluation are shared (Methods). Loss, optimizer, learning rate, weight decay, schedule shape and effective batch are taken from each model's source recipe; the schedule is stretched over the shared number of steps. Models with a fixed input size train on patches of that size and predict with half-overlapping windows; all others train on 960×960 patches and predict on the whole canvas. nnU-Net runs through its own entry point and keeps its own plan.

Model	Initialization	Head	Loss	Optim.	LR	WD	Schedule	Batch	Patch	Inference	Steps
U-Net	none	4 sigmoid	BCE + Dice[a]	Adam	1e-4	0	constant	4	960	whole canvas	21,000
DeepLabv3	ImageNet, ResNet-101	4 sigmoid	BCE	SGD	7e-3	1e-4	poly 0.9	16	960	whole canvas	6,000
HRNet-W48	ImageNet	4 sigmoid	BCE	SGD	1e-2	5e-4	poly	12	960	whole canvas	7,000
Swin-Unet	ImageNet, Swin-T	5 softmax	0.4 CE + 0.6 Dice	SGD	1e-2	1e-4	poly 0.9	24	224	sliding window	63,000
FCT	none	4 sigmoid	BCE	Adam	1e-3	0	constant	4	384	sliding window	129,000
nnU-Net	none	5 softmax	CE + batch Dice	SGD	1e-2	3e-5	poly 0.9	2	1024×1536[c]	sliding window	250,000[c]
nnU-Net ResEnc-M	none	5 softmax	CE + batch Dice	SGD	1e-2	3e-5	poly 0.9	13[c]	384×640[c]	sliding window	250,000[c]
M2MRF	ImageNet, HRNet-W48	4 sigmoid	binary Dice	SGD	1e-2	5e-4	poly 0.9	4	960	whole canvas	21,000
HACDR-Net	none	5 softmax	weighted CE[b]	AdamW	6e-5	1e-2	poly 1.0, warm-up	1	960	whole canvas	83,000
HRDecoder	ImageNet, HRNet-W48	4 sigmoid	binary Dice + 0.1 Dice (HR crops)	SGD	1e-2	5e-4	poly 0.9	4	960	sliding window	21,000
PMCNet	ImageNet, EfficientNet-B0	5 softmax	CE	Adam	1e-4	0	poly 0.9	1	960	whole canvas	83,000
WSRFNet	ImageNet, EfficientNet-B1	4 sigmoid	binary Dice, per loop[d]	SGD	1e-2	5e-4	poly 0.9	4	960	whole canvas	21,000
CLC-Net[C]	none	5 softmax	Dice + BCE (cls. heads)[e]	Adam	3e-4	0	step (1/3, 5/6)	16	512	sliding window	73,000
TC-Net[C]	none	5 softmax	CE	Adam	5e-4	1e-5	constant	4	512	sliding window	73,000
SAMed	SAM ViT-B, LoRA r=4	5 softmax	0.2 CE + 0.8 Dice	AdamW	5e-3	0.1	warm-up, poly	24	512	sliding window	13,000
SAM-Adapter	SAM ViT-H, adapters	4 sigmoid	BCE + IoU	AdamW	2e-4	1e-2	stepped cosine	2	1,024	sliding window	37,000
SAM2-UNet	SAM 2 Hiera-L, adapters	4 sigmoid	BCE + weighted IoU[f]	AdamW	1e-3	5e-4	stepped cosine	12	960	whole canvas	7,000
MedSAM (full FT)	MedSAM ViT-B, all weights	4 sigmoid	Dice + BCE	AdamW	1e-4	1e-2	constant	2	1,024	sliding window	37,000
RETFound + UPerNet	RETFound ViT-L (MAE)	4 sigmoid	BCE + 0.4 aux.	AdamW	4.7e-4	0.05	warm-up, cosine	24	224	sliding window	63,000

Batch is the effective batch (gradient accumulation where one card cannot hold it). Steps is for IDRiD at 1,000 equivalent epochs; one equivalent epoch draws as many patch pixels as the training canvases contain, so steps = $\lceil \lceil$canvas pixels / patch pixels$\rceil$ / batch$\rceil$ per epoch. DDR uses 400 equivalent epochs (for example 83,600 steps at batch 4 and 960-pixel patches; Supplementary Table 3 lists every run). [C], re-implemented from the paper. Every run's recipe record, with its deviations from the source recipe, is kept in the code repository (to be released).
[a]Per-image, per-class sigmoid Dice with smoothing 1 in the registered recipe; the main-table runs pool the Dice term over the batch (Methods). The other Dice losses use 10^{-5}. [b]IDRiD: class-weighted cross-entropy; DDR: the same plus the authors' NALoss with weight 0.2. [c]nnU-Net plans patch and batch per dataset (IDRiD plans shown; DDR: default 1024×1536 at batch 2, ResEnc-M 512×512 at batch 12) and trains 1,000 epochs of 250 steps on both sources; its learning rate, weight decay and polynomial schedule are its defaults. [d]The paper's loss and optimizer, written into the wrapper because the repository has no training loop. [e]Local and contextual Dice plus binary cross-entropy of the two classification heads, as described in the paper. [f]Registered per-image term; the main-table run pools it over the batch (Methods).

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